# Beyond the average phenotype: individual variation alters mechanistic forecasts of climate responses

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RUNNING TITLE: Individual variation alters forecasts

**Abstract**

Mechanistic models are central to forecasting biological responses to climate change because they simulate organism–environment interactions that determine growth, survival and reproduction. Yet, these models generally represent species by a single average phenotype despite evidence of substantial physiological variation among individuals. Because organismal performance emerges from nonlinear interactions, Jensen's inequality predicts that forecasts based on average phenotypes may differ from those generated by heterogeneous populations. Whether this principle influences mechanistic climate forecasts remains largely unknown. Here, we combined longitudinal measurements of growth and metabolism from individually monitored alpine newts (*Ichthyosaura alpestris*) with Dynamic Energy Budget (DEB) models to estimate physiological parameters at the species and individual levels. We coupled these parameterisations with biophysical models to predict twelve life-history and physiological performance traits across Europe under current climate and +2 °C and +4 °C warming scenarios, and compared species- and individual-level forecasts to quantify Jensen bias. Forecasts based on the average phenotype consistently diverged from those generated by heterogeneous individuals, demonstrating that physiological averaging introduces systematic bias into mechanistic ecological forecasts. Bias magnitude, direction and, in some cases, even sign depended on the performance trait, varied geographically, and changed under climate warming. Reproductive traits were consistently more sensitive to physiological averaging than structural traits, whereas warming frequently amplified, attenuated or reversed forecasting bias depending on trait and environmental context. Despite these differences, both parameterisations recovered the same broad spatial pattern of biological responses to climate warming, but differed in the magnitude of the predicted changes. Our results identify biological heterogeneity as an important source of uncertainty in mechanistic climate forecasting. Representing populations as distributions of physiological phenotypes, rather than single average individuals, provides a practical path towards forecasts that quantify not only expected ecological responses to climate change, but also where those predictions are robust and where they are intrinsically susceptible to bias arising from nonlinear averaging.

**Keywords:** biophysical modeling; climate change; Dynamic Energy Budget theory; intraspecific variation; Jensen's inequality; niche modeling

# 1 Introduction

Predicting how populations will respond to climate change remains one of the central challenges in ecology. Increasingly, these predictions rely on mechanistic approaches that simulate how organisms interact with their ~~environments~~abiotic environment, thus, linking environmental conditions to organismal performance, including growth, survival and reproduction (Briscoe et al., 2023; Kearney & Porter, 2009; Kearney & Porter, 2017, 2020). By coupling organismal functional traits (e.g., morphology, physiology and behaviour) with microclimatic conditions derived from first principles, mechanistic models generate biologically grounded forecasts, overcoming some of the limitations of purely correlative approaches (Briscoe et al., 2023; Kearney & Porter, 2009; Urban et al., 2016), especially when environmental conditions extend beyond those currently experienced by species (Elith & Leathwick, 2009; Maino et al., 2016). As a consequence, climate forecasting has become increasingly realistic in its representation of environmental heterogeneity. Yet, species or populations are almost invariably represented by a single set of physiological parameters corresponding to an average individual (but see Enriquez-Urzelai et al., 2020; Kolbe et al., 2010; Riddell et al., 2022). Whether this simplification adequately captures population responses to environmental change remains largely unknown.

This assumption contrasts with a growing recognition that variation among individuals is a fundamental component of ecological systems. Individuals within populations frequently differ in morphology, physiology, behaviour and life-history traits, and these differences influence resource use, species interactions, demographic rates and ecosystem functioning (Bolnick et al., 2003, 2011; Des Roches et al., 2018). Indeed, intraspecific variation can rival or exceed interspecific variation in its ecological effects (Des Roches et al., 2018), prompting increasing recognition that populations are better viewed as distributions of phenotypes rather than collections of equivalent individuals (Grimm et al., 2025; Jeltsch et al., 2025; Moran et al., 2016). Among the many dimensions of individual variation, physiology is particularly relevant for understanding responses to environmental change because it provides the mechanistic link between environmental conditions and ecological performance (Seebacher & Franklin, 2012). Physiological traits determine how organisms interact with their thermal ~~, hydric and energetic environments~~ and hydric environments and acquire and allocate energy, thereby shaping growth, reproduction, survival and, ultimately, fitness. Individuals commonly differ in metabolic rates, thermal sensitivities, or energy allocation and the plasticity of these traits ~~(Dall et al., 2012; Gvoždík, 2024; Pettersen & Metcalfe, 2024)~~ (Gvoždík, 2024; Pettersen & Metcalfe, 2024). Such physiological diversity means that individuals experiencing the same environmental conditions may differ in their performance and, consequently, may contribute differently to population responses under environmental change. As a result, climate change acts on distributions of heterogeneous phenotypes, whereas mechanistic forecasts commonly represent populations using a single average phenotype.

The ecological consequences of physiological variation (e.g., consistent among individual variation in physiological traits), however, cannot be understood from trait variation alone. Performance emerges from the interaction between organismal phenotypes and their environments through multiple interacting biological processes. These interactions can be conceptualized as multidimensional performance surfaces, which describe how performance varies across combinations of organismal phenotypes and environmental (abiotic) conditions (Laughlin and Messier, 2015; Li et al., 2021; **Fig. 1a**). Such surfaces

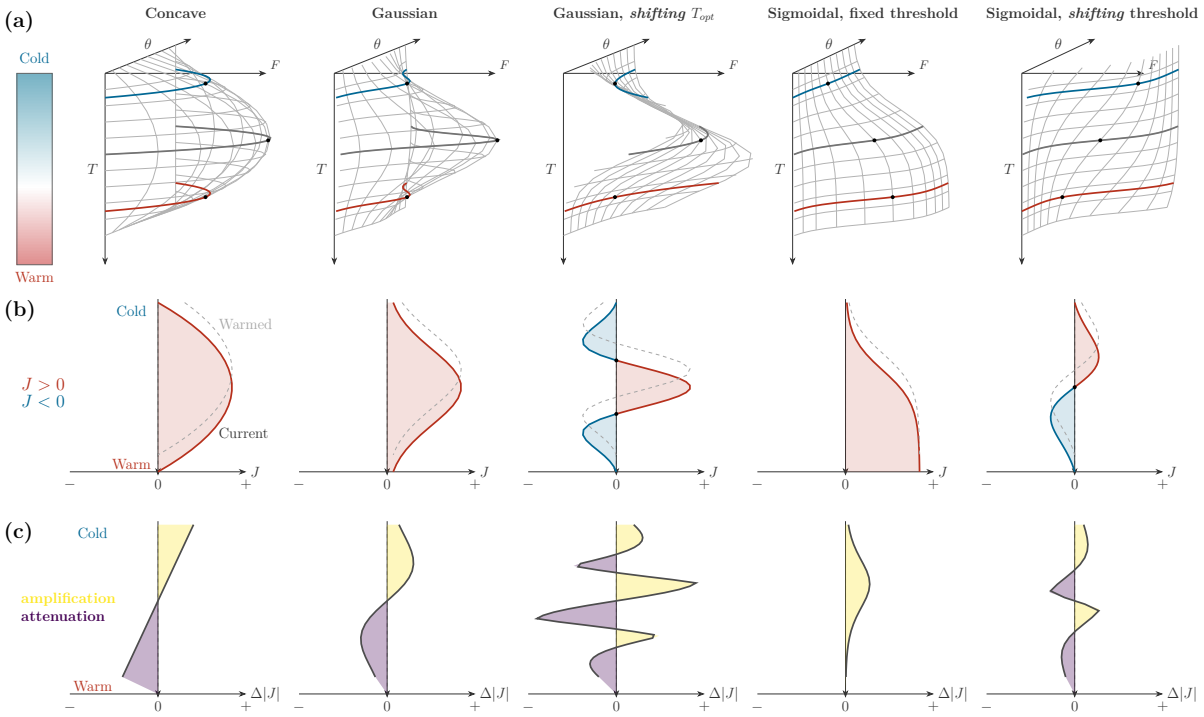


**Figure 1:** Jensen's inequality and its implications for forecasting biological responses under climate change. For a nonlinear function the value at the mean $f(\bar{\theta})$ differs from the mean of the values $\overline{f(\theta)}$, producing a forecasting bias (hereafter Jensen bias, $J = f(\bar{\theta}) - \overline{f(\theta)}$), where $J \approx -\frac{1}{2} f''(\bar{\theta})\,\mathrm{Var}[\theta]$, and whose sign and magnitude follow the local curvature of $f$ (Denny, 2017; Jensen, 1906). **(a) Representative trait–performance surface illustrating common forms of nonlinear responses to temperature**. The blue-to-red bar on the far left represents an environmental temperature gradient (cold to warm). Performance ($F$) is shown as a function of trait value ($\theta$) and temperature ($T$) for five illustrative types: concave dome, Gaussian, Gaussian with shifting thermal optimum ($T_{opt}$), sigmoidal with fixed threshold, and sigmoidal with shifting threshold. Colored vertical slices show the trait–performance relationship at cold (*blue*), intermediate (*black*) and warm (*red*) temperatures, while *black points* denote the mean trait value. **(b) Temperature-dependent Jensen bias predicted for each response surface**. *Red shading* indicates positive bias ($J > 0$), where predictions based on the mean trait overestimate performance, whereas the *blue shading* indicates negative bias ($J < 0$), where the mean trait underestimates performance. The *solid curve* represents the bias under the current climate, $J(T)$, and the *dashed curve* the bias after warming, $J(T + \Delta T)$, illustrating how climate change shifts the magnitude and sign of forecasting bias for a fixed trait distribution after warming ($\bar{\theta}$ and $\mathrm{Var}[\theta]$ held constant). **(c) Impact of climate warming on the magnitude of Jensen bias**. The *black curve* shows the change in absolute bias, $|J|$, across the temperature gradient. *Yellow shading* denotes amplification ($|J|$ increases), whereas *purple shading* denotes attenuation ($|J|$ decreases). Together, the examples demonstrate that the same warming scenario can either strengthen or weaken forecasting bias depending solely on the curvature and temperature dependence of the underlying trait–performance relationship.

are rarely linear. Instead, they commonly exhibit thermal optima, thresholds, asymmetries and other nonlinear features that are pervasive in ecological systems (Bolnick et al., 2011; Ruel & Ayres, 1999). Consequently, the ecological consequences of individual variation depend not only on the amount of physiological variation within a population, but also on the environmental context in which it is expressed: the same distribution of physiological phenotypes may produce similar performance in some environments but markedly different performance in others (Ross et al., 2023). Moreover, different components of performance are likely to be associated with different performance surfaces. Growth, reproduction or survival might not share the same thermal optima, thresholds or nonlinearities (Kontopoulos et al., 2024; Sinclair et al., 2016), suggesting that the effects

of physiological variation, and the biases introduced by physiological averaging, may differ among components of organismal performance.

These ideas have important consequences for ecological forecasting. When the relationship between phenotype, environment and performance is ~~not proportional (i.e., it is nonlinear)~~nonlinear, relying on an average phenotype for predictions yields different results than accounting for the phenotypic diversity present in real populations, a consequence of Jensen's inequality (Jensen, 1906; Ruel and Ayres, 1999; **Fig. 1a,b**). Importantly, this mismatch, or Jensen bias $J$, does not arise from physiological variation alone, but from ~~the interaction between the distribution of phenotypes and~~ its interaction with the geometry of the underlying performance surface (Denny, 2017, 2019; Laughlin & Messier, 2015)~~.~~ ~~Where~~ : where the local curvature ~~of the performance surface~~ is convex, predictions based on the average phenotype underestimate the average performance of heterogeneous populations ($J < 0$)~~; where the curvature~~ , whereas where it is concave, ~~predictions based on the average phenotype overestimate the average performance of heterogeneous populations~~ they overestimate it ($J > 0$; Bolnick et al., 2011; Ruel and Ayres, 1999). ~~Consequently, both the magnitude and the direction of the Jensen bias depend on the local geometry of the performance surface (**Fig. 1b**).~~ Different functional forms (e.g. concave, Gaussian and threshold-like relationships), as well as different regions within the same performance surface (e.g. along environmental gradients or across the geographic distribution of a species), therefore generate distinct patterns of averaging bias ~~. As illustrated in **Fig. 1c**,~~ (**Fig. 1b**). Because climate change alters ~~the~~ environmental conditions (Simon & Amarasekare, 2024) and ~~shift~~ thus where populations lie on the performance surface~~across the distribution range of species. As a consequence, the mismatch between forecasts generated from average phenotypes and those generated from heterogeneous populations~~ , Jensen bias may increase (amplify), decrease (attenuate) or even reverse ~~depending on how environmental change reshapes the local geometry of the performance surface. Characterising how Jensen bias varies across traits, environmental gradients and climate scenarios is therefore essential for understanding when and where physiological averaging matters most for ecological forecasts~~under warming (**Fig. 1c**).

Here, we combined longitudinal measurements of growth and metabolic rates from individually monitored alpine newts (*Ichthyosaura alpestris*) ~~(Baškiera & Gvoždík, 2020)~~ (Baškiera & Gvoždík, 2021) with mechanistic modeling: Dynamic Energy Budget, DEB, and biophysical models in particular. This framework allowed us to evaluate how consistent individual physiological variation (i.e., non-plastic physiological variation with a genetic and developmental basis) influences climate-change forecasts across the species' geographic distribution. Because DEB theory mechanistically links environmental conditions to energy acquisition, allocation and expenditure throughout the life cycle (Kooijman, 2010), and coupled biophysical models reconstruct the thermal and hydric environments experienced by organisms from local microclimates (Kearney & Porter, 2009; Kearney et al., 2013), this framework is well-suited to evaluate the ecological consequences of Jensen's inequality. Growth, reproduction, survival and other life-history traits emerge from nonlinear interactions among physiology, energetics and environmental conditions. Thus, empirically measured physiological variation naturally propagates through realistic trait–environment performance surfaces under current and future climates (**Fig. 1**). By comparing forecasts generated from species-level and individual-level physiological parameterizations under identical environmental simulations, we directly quantified the forecasting bias introduced by physiological averaging. Specifically, we tested 1) whether forecasts based on an average physiological traits diverge from those obtained by propagating empirically

measured individual-level physiological variation; 2) whether the magnitude and direction of Jensen bias vary among performance traits and across environmental gradients; and 3) whether climate change reshapes these patterns by amplifying, attenuating or reversing forecasting bias. By linking empirical estimates of individual physiological variation to the geometry of trait–environment performance surfaces, this study quantifies when and where physiological averaging matters most for ecological forecasts under climate change.

# 2 Materials and Methods

## 2.1 Study system and data

We used the alpine newt (*I. alpestris*) as a model species, a temperate amphibian widely distributed across Europe. Juveniles are ~~entirely~~ strictly terrestrial: they usually leave their aquatic larval habitat and disperse on land until sexual ~~maturation~~ maturity (Denoël, 2004). Adults spend most of the year on land except ~~during~~ the breeding season (April–June). Terrestrial individuals are predominantly nocturnal or crepuscular and use a variety of daytime retreats, including leaf litter, stones, dead tree trunks and burrows of other animals (Roček et al., 2003). Juveniles, as other predominantly nocturnal amphibians (Enriquez-Urzelai et al., 2020), can show diurnal activity (specially after metamorphosis) to acquire energy reserves before their first winter. The preferred body temperature of the species is 17.3 °C, with a lower bound of thermal preference of 13.8 °C and an upper bound of 20.0 °C (Gvoždík & Kristín, 2017), although ~~activity may occur at temperatures as low as~~ individuals remain capable of locomotion at temperatures approaching 1 °C.

We compiled two complementary datasets describing the life history and physiological variation of the species. The first dataset consisted of species-level life-history and growth information: these data included lifespan (20 years), snout–vent length at metamorphosis (1.83 cm), ~~snout–vent length at~~ female and male ~~puberty~~ snout–vent length at sexual maturity (4.9 and 4.4 cm, respectively), ~~ultimate body size~~ asymptotic snout–vent length (5.84 cm females, 5.05 cm males), ~~ultimate wet weight~~ asymptotic body mass (3.16 g females, 2.04 g males), and maximum reproductive rate (600 eggs $\text{year}^{-1}$) (Gvoždík & Kristín, 2017; Miaud et al., 2000). Species-level information was complemented with published growth (time–snout-vent-length) trajectories (Miaud et al., 2000). The second dataset consisted of longitudinal measurements from 37 alpine newts (18 females and 19 males) from a population near Jihlava, Czech Republic (49.2247° N, 16.0579° E; 600 m a.s.l.). These individuals were monitored annually from 2014 (year of metamorphosis) to 2018. Each year, body mass (in g) and standard metabolic rate (~~oxygen consumption at rest~~ in mL $O_2$ $\text{h}^{-1}$) were measured, providing growth and metabolic trajectories for each individual ~~(Baškiera & Gvoždík, 2020)~~(Baškiera & Gvoždík, 2021).

## 2.2 DEB model parameter estimation

Dynamic Energy Budget (DEB) theory (Kooijman, 2010; Sousa et al., 2008) describes the acquisition, storage, and allocation of energy and mass throughout the life cycle. The standard DEB model (Marques et al., 2018) represents an organism by three state variables: structural volume $V$, reserve energy $E$, and cumulative energy invested in maturation $E_H$ (plus a reproduction buffer $E_R$ in adults). Energy ~~enters the organism as food,~~ is assimilated into the reserve at a rate proportional to surface area ~~, and is subsequently mobilised (catabolised) from the reserve to fuel five downstream processes:~~ and mobilised from it to

fuel somatic maintenance, growth~~in structure~~, maturity maintenance, and maturation (in juveniles) ~~, and~~ or reproduction (in adults)~~. Reproduction energy accumulates in a buffer and is periodically converted into eggs. Life-history transitions occur at species-specific maturity thresholds: ~~ , with life-history transitions occurring at maturity thresholds, e.g., birth at $E_H^b$ and sexual maturity at $E_H^p$ ~~(Kooijman, 2010; Marques et al., 2018)~~.~~

~~The behaviour of the standard~~ (Kooijman, 2010; Marques et al., 2018; Oliveira et al., 2026) . The model is governed by 12 primary parameters (Kooijman, 2010). ~~The maximum surface-specific assimilation rate $\{\dot{p}_{Am}\}$ controls the rate of energy acquisition and, through covariation rules, scales with maximum body size via the dimensionless~~ Among them, the zoom factor $z$ ~~. A larger $z$ implies a larger body , a proportionally higher assimilation capacity, and a greater energetic investment at life-history transitions (Sousa et al., 2008). Once assimilated,~~ scales maximum body size and the maximum surface-specific assimilation rate $\{\dot{p}_{Am}\}$ (Sousa et al., 2008); reserves are mobilised at a rate set by the energy conductance $\dot{v}$~~and~~ ; and, following the $\kappa$-rule~~: a fixed~~ , a fraction $\kappa$ of the mobilised reserve is allocated to somatic maintenance, ~~whose cost per unit structural volume is $[\dot{p}_M]$, and to growth, at a specific cost of $[E_G]$ per unit of new structure; the remainder $1-\kappa$ supports maturity maintenance (at a rate set by $\dot{k}_J$) and maturation in juveniles or reproductionin adults, with eggs produced at an efficiency $\kappa_R$. The rates of feeding and assimilation are further determined by the maximum specific searching rate $\{\dot{F}_m\}$, the assimilation efficiency $\kappa_X$, and the surface-specific maintenance cost $\{\dot{p}_T\}$, which is assumed to be zero for ectotherms. The cumulative energetic investments at which organisms reach birth and sexual maturity are defined by the thresholds $E_H^b$~~ at a cost $[\dot{p}_M]$ per unit of structural volume, and ~~$E_H^p$. In addition, auxiliary parameters include the shape coefficient ($\delta_M$, which allows to convert shape-specific length measures to lengths relevant in the DEB context) and the acceleration of ageing ($\ddot{h}_a$). In addition, to account for between-sex differences, sex-specific parameters can be used: e.g., for male puberty ($E_{H,m}^p$) and body-size dimorphism ($z_m$). In species with a larval stage, the standard model can be further extended with an additional maturity threshold at metamorphosis $E_H^j$~~growth, while the remainder supports maturation or reproduction. Because each parameter quantifies the rate or efficiency of a specific physiological process, the ~~standard DEB~~ model provides a coherent framework for interpreting individual variation in terms of energetic traits with distinct ecological meaning (Kooijman et al., 2021). A full description of the model and its parameters is given in **Appendix 1**.

Because DEB parameters describe latent energetic processes rather than directly measurable biological traits, they were estimated by inverse modelling using the covariation method (Lika et al., 2011; Marques et al., 2018)~~. We fitted the standard DEB model for *I. alpestris* using MATLAB (The MathWorks Inc.) together with the freely available~~ , with the AmPtool and DEBtool_M packages in MATLAB (The MathWorks Inc.). Parameter estimation was performed at two levels: first, a single species-level parameterization representing the average physiological phenotype (i.e., as is standard practice in mechanistic modelling, a single parameter set fitted to data pooled from multiple individuals; hereafter, average phenotype), and second, separate parameterizations for each of the 37 individuals. ~~Model goodness-of-fit was assessed using the mean relative error (MRE) and symmetric mean squared error (SMSE) statistics (Marques et al., 2018) .~~

For the species-level parameterization, the model was fitted ~~using~~ simultaneously to the species-level life-history information ~~together with the pooled~~ and the complete set of longitudinal growth and metabolic ~~trajectories~~ observations from all 37 individu-

als, which were retained as separate data points rather than averaged across individuals or ages. Of the 12 primary parameters~~that govern the standard model, seven were estimated : the maximum surface-specific assimilation rate $\{\dot{p}_{Am}\}$, via the zoom factor~~, we estimated seven ($z$~~; energy conductance~~, $\dot{v}$~~; the allocation fraction~~, $\kappa$~~; volume-specific somatic maintenance~~, $[\dot{p}_M]$~~;~~, the specific cost of structure $[E_G]$~~;~~, and the maturity thresholds ~~at birth~~ $E_H^b$ and ~~puberty~~ $E_H^p$~~. The remaining five primary parameters ($\{\dot{F}_m\}$, $\kappa_X$, $\kappa_R$), $\{\dot{p}_T\}$ and $\dot{k}_J$)were fixed to standard values from the DEB literature (Kearney et al., 2013; Kooijman, 2010). In addition, we estimated a few species-specific additional parameters: the~~ together with a maturity threshold at metamorphosis ~~$E_H^j$;~~ ~~the male puberty threshold $E_{H,m}^p$ and male zoom factor~~ and male-specific zoom factor ($z_m$~~,~~) and puberty threshold ($E_{H,m}^p$) to account for sexual size dimorphism; ~~and the shape coefficient ($\delta_M$) and Weibull ageing acceleration ($\ddot{h}_a$) parameters. The Arrhenius temperature $T_A$ and all chemical~~ the remaining parameters were fixed at standard values ~~. Model parameters (~~(Kearney et al., 2013; Kooijman, 2010). Parameter values are given in **Table S2.1**~~) and goodness of fit can be found at . Together, these estimated parameters define the physiological trait space of the species, with each parameter controlling a distinct aspect of energy acquisition, allocation, or expenditure~~, and further details of the estimation and goodness of fit in **Appendix 1**.

Individual-level parameters (**Table S2.1**) were obtained by re-estimating ~~a subset of four~~ three parameters independently for each individual: the sex-specific zoom factor ($z$ for females, $z_m$ for males), energy conductance ($\dot{v}$), and volume-specific somatic maintenance ($[\dot{p}_M]$). We allowed these parameters to vary among individuals because they govern the processes captured by our individual-level data: variation in body size and energy uptake (via $z$ and $z_m$), in energy mobilisation (via $\dot{v}$), and in maintenance costs (via $[\dot{p}_M]$) jointly determine how individuals acquire and use energy, and thus how they grow and respire under a given environment (Kooijman et al., 2021). All remaining parameters, including $\kappa$, $[E_G]$, and the maturity thresholds, were held at their species-level values. The estimation used each individual's own longitudinal growth and metabolic trajectories as univariate data, with species-level estimates serving as starting values. Each individual was assigned the sex-appropriate puberty threshold ($E_H^p$ for females, $E_{H,m}^p$ for males). Consequently, differences among individuals in these ~~four parameters represent~~ parameters can be interpreted as functionally coherent physiological variation~~: variation in body size and energy uptake (via $z$ and $z_m$), in energy mobilisation (via $\dot{v}$), and in maintenance costs (via $[\dot{p}_M]$) jointly determine how individuals acquire and use energy, and grow under a given environment (Kooijman et al., 2021)~~.

## 2.3 Mechanistic modeling framework

To simulate organismal performance under current and future climates, we coupled biophysical and Dynamic Energy Budget (DEB) models using the NicheMapR R package (Kearney & Porter, 2017, 2020). This framework uses biophysical models to compute hourly body temperature ($T_b$), activity windows, and foraging opportunities given available microclimates and the organism's morphological, physiological and behavioural traits (Kearney et al., 2018; Porter et al., 1973). These hourly values drive the DEB model: $T_b$ is applied to all metabolic rates through the Arrhenius correction, and, since we held food availability constant through the simulations, the time available for foraging determines the amount of energy (food) assimilated, which, together with body temperatures, shapes how organisms grow, develop and reproduce (Kearney, 2012; Kearney & Porter, 2020).

We reconstructed microclimatic conditions ~~using~~ with NicheMapR's microclimate model, as implemented in the `micro_global()` function (Kearney & Porter, 2017; Kearney et al., 2014). The model uses monthly climatological ~~data (maximum and minimum air temperature, precipitation, cloud cover, relative humidity, wind speed and number of wet days)~~ means from the global 10 arc-minute climatology of New et al. (2002), together with ~~elevation, slope, aspect~~ terrain and soil properties, to estimate hourly ~~aboveground~~ air temperature, ~~solar and infrared~~ radiation, wind speed and relative humidity at the animal's height, as well as soil temperature and moisture profiles ~~at ten depths~~ (Kearney & Porter, 2017; Kearney et al., 2014). Microclimate simulations were run for ~~23~~ 22 years with the moisture and snow routines enabled (Kearney, 2020; Kearney & Maino, 2018)~~.~~ ~~The~~ , excluding the first two years ~~were excluded~~ as spin-up~~to ensure stabilized snow and soil moisture dynamics before organismal simulations . We run~~ , which resulted in simulations of up to 20 years (i.e., the maximum known lifespan of the species). We ran the microclimate simulations ~~in two shading conditions:~~ under fully unshaded (0% shade) and deep shade (90% shade) conditions.

To estimate *I. alpestris* performance and simulate its life cycle~~with coupled biophysical and DEB models~~, we used the `ectotherm()` function of NicheMapR (Kearney & Porter, 2020)~~. We parameterised the ectotherm model~~ , parameterised for a terrestrial alpine newt ~~,~~ with 90% of the skin surface acting as a free water exchanger ~~to account for evaporative water loss~~ (Enriquez-Urzelai & Gvoždík, 2024). Individuals were assumed to be nocturnal and crepuscular, but juveniles were also allowed diurnal activity ~~, reflecting the need~~ to acquire energy reserves before their first winter (Enriquez-Urzelai et al., 2020). Animals ~~retreated underground during daylight hours (in the case of adults), and whenever body temperature fell outside the~~ were active only when body temperature was within their voluntary thermal limits ($VT_{min}$ = 4 °C, $VT_{max}$ = 20 °C) ~~, or relative humidity aboveground was lower than~~ and aboveground relative humidity was at least 65% ~~. When burrowed, animals selected a depth (from~~ (Bonnefond et al., 2020; Giacometti & Tattersall, 2025); otherwise, they retreated underground, selecting depths (2.5 cm ~~up~~ to 2 m) that ~~prevented them from reaching~~ kept them within their critical thermal limits ($CT_{min}$ = −2 °C~~and~~ , $CT_{max}$ = 36 °C) (Enriquez-Urzelai & Gvoždík, 2024). ~~The preferred body temperature was set to (Gvoždík & Kristín, 2017). Although amphibians are not as accurate thermoregulators as e.g. reptiles, we allowed individuals to behaviourally~~ When active, individuals could select shaded microhabitats ~~to stay within their preferred temperature range (Enriquez-Urzelai et al., 2019). In the case of females (both at the species and individual level), we simulated reproduction using~~ (0–90% shade) to stay close to their preferred body temperature of 17.3 °C (Enriquez-Urzelai et al., 2019; Gvoždík & Kristín, 2017) . Females reproduced following the batch reproductive model (Pecquerie et al., 2009)~~:~~ ~~females stored energy in the reproductive buffer until it was enough to produce a clutch (200-600 eggs) during the breading season. Alternatively, we simulate males to continuously deplete~~ , producing clutches of 200–600 eggs, whereas in males the reproductive buffer ~~during the breading season, so no energy allocated to reproduction was stored. The breading season occurred~~ was continuously depleted during the breeding season (between the spring equinox and the summer solstice~~. We run the simulations from~~ ). Simulations started at the juvenile stage ~~(starting from the~~ on 15 ~~of August , when juveniles are present in the field), for the total length of the microclimate simulation, until death happened~~August and ran until the individual died (see **Appendix 1** for further details).

Forecasts were generated for every ~~grid cell within Europe, using~~ terrestrial cell of a 10 arc-minute grid covering Europe (10° W–40° E, 35°–70° N). For each grid cell, the coupled ectotherm–DEB model was run using the species-level parameterisation (female and male) and each of the 37 individual parameterisations (18 females and 19 males). The species-level and individual-level DEB parameter sets included the appropriate sex-specific zoom factors ($z$, $z_m$) and puberty thresholds ($E_H^p$, $E_{H,m}^p$).

To evaluate the thermal effects of climate change, the complete simulation framework was repeated under two warming scenarios corresponding to uniform increases of +2 °C and +4 °C in hourly air temperature relative to the current climate. These scenarios encompass the range of projected warming across Europe under intermediate to high emission pathways (Calvin et al., 2023). Warming was implemented as a uniform additive offset to ~~hourly air temperature generated by the microclimate model~~the baseline climate's temperature forcing variable, whereas all other climatic variables (precipitation, relative humidity, wind speed, cloud cover and solar radiation) remained unchanged. This approach isolates the direct thermal effects of warming on ectotherm energetics while allowing direct comparison among climate scenarios. Note, however, that because relative humidity was held constant, warmer air implies a higher vapour pressure deficit and, therefore, higher evaporative water loss.

## 2.4 Quantifying Jensen bias

We quantified Jensen bias as the difference between the prediction obtained from the average phenotype (i.e., the species-level parameterisation) and the mean of the predictions obtained from the individual phenotypes (i.e., the individual-level parameterisations). For every grid cell, physiological parameterisation and climate scenario, we extracted 12 model outputs describing life-history and physiological performance: lifespan (year), lifetime ~~fecundity~~ reproductive output (number of eggs in females; reproductive energy, J, in males), net reproductive rate ($R_0$), maximum and minimum annual body temperatures ($T_{b,max}$, $T_{b,min}$), maximum body ~~weight~~ mass (g), maximum ~~body~~ snout–vent length (cm), minimum reserve energy (~~Joule~~J), cumulative basking time (h), cumulative foraging time (h), cumulative food intake (g, dry mass) and mean daily water need (g day$^{-1}$). Cumulative metrics were summed over each individual's entire simulated lifetime.

For each performance metric, we generated four prediction maps: two species-level predictions ($P_{\text{species}}$, for females and males) obtained from the species-level parameterisation, and two individual-level ~~prediction~~ predictions obtained by averaging the outputs from the individual parameterisations ($\overline{P}_{\text{ind}}$, for female and male individuals). Forecasting bias associated with physiological averaging was quantified for every grid cell as

$$J = P_{\text{species}} - \overline{P}_{\text{ind}}$$

where $J$ measures the discrepancy between the species-level prediction and the mean of the individual-level predictions. Positive values of $J$ indicate that the species-level parameterisation overpredicts performance relative to the mean of individuals, whereas negative values indicate underprediction. We computed $J$ for females and males separately. Because performance metrics differ in their units, we compared the magnitude of bias among metrics using the normalised absolute Jensen bias, i.e., the mean $|J|$ across grid cells divided by the mean $|\overline{P}_{\text{ind}}|$ across grid cells (×100).

To quantify spatial variation in the ~~ecological consequences of physiological heterogeneity~~ relative among-individual variability of predicted performance, independently of Jensen bias, we also calculated the coefficient of variation (~~*CV*~~CV) across the ~~37~~ individual-level predictions ~~for each sex ,~~ of each sex (18 females and 19 males), for each metric and grid cell~~. High *CV* ~~, and compared its median among southern (<40° N), central and northern (>60° N) latitudes. High CV values identify geographical regions and physiological metrics where individual ~~variation most strongly influences predicted outcomes, highlighting areas in which species-average forecasts may be particularly misleading~~differences translate into the most divergent predicted outcomes.

To evaluate how climate warming modified forecasting bias, we calculated the absolute Jensen ~~dissimilarity~~ bias ($|J|$) for each climate scenario and computed

$$\Delta|J| = |J_{\text{future}}| - |J_{\text{current}}|$$

Positive values indicate amplification of forecasting bias under warming, whereas negative values indicate attenuation. ~~Paired Wilcoxon signed-rank tests (one per metricacross all grid cells ; two-sided alternative)were used to compare $|J|$ between current and future climates~~For each metric, we summarised the change in bias as the relative change in absolute Jensen bias (rel. $\Delta J$), i.e., the mean $\Delta|J|$ across grid cells divided by the mean $|J_{\text{current}}|$ across grid cells ($\times 100$). We additionally quantified the proportion of grid cells in which $|J|$ increased, decreased or remained unchanged, as well as the proportion in which the sign of $J$ (overprediction versus underprediction) reversed between climate scenarios.

## 2.5 Consequences of Jensen bias for forecasting

To examine whether Jensen bias varied across environmental gradients, we related the change in absolute Jensen bias ($\Delta|J|$) under ~~warming scenarios~~ +4 °C warming to mean annual air temperature ($\overline{T}_a$)~~. $\overline{T}_a$ was ~~, calculated for each grid cell from the ~~global 10 arc-minute~~ climatology of New et al. (2002)~~by averaging the monthly means of the daily maximum and minimum air temperatures. The~~ . For each metric and sex, $\Delta|J|$ ~~values were rasterized at approximately 0.16° resolution for each of the 12 performance metrics and summarized~~ was summarised as binned means ~~(±1 SE)~~ in 1 °C $\overline{T}_a$ intervals, ~~retaining only bins containing at least 10 grid cells. Generalised additive models (GAMs) with cubic regression splines were fitted separately for each metric, and predictions with 95% confidence intervals were evaluated at 0.1 °C intervals along the~~ and the overall trend was described with a generalised additive model (GAM) with a smooth term for $\overline{T}_a$ ~~gradient~~using the `mgcv` R package (Wood, 2025). Because model outputs are deterministic and neighbouring grid cells are spatially autocorrelated, GAMs were used as descriptive smoothers rather than for statistical inference (model specification in **Appendix 1**).

To evaluate the ecological implications of physiological averaging more directly, we compared forecasts generated from the species-level and individual-level physiological parameterisations for three representative life-history traits, summarising survival, growth and reproduction: lifespan, maximum body mass and lifetime fecundity (or reproductive energy for males). ~~These traits summarize major fitness/performance components: survival, growth and reproduction.~~ For each trait, grid cell and climate scenario, we ~~extracted~~ compared the species-level prediction ~~($P_{\text{species}}$) and~~ with the corresponding mean individual-level prediction~~($\overline{P}_{\text{ind}}$). Comparisons included both the absolute predictions~~ , both under the current climate and ~~the~~ as projected changes under warming ~~scenarios. Future~~

projected changes were expressed relative to the current climate ($P_{\text{future}} - P_{\text{current}}$)for each physiological parameterisation. For visual purposes, we normalized the projected changes relative to the current climate to range between $[-1, 1]$ by dividing each grid cell's value by the maximum absolute changewithin each trait. We report the percentage of grid cells in which the species-level model under- or overpredicted the individual-level mean, and the percentage in which both parameterisations projected the same direction of change. In addition, we computed the relative biasfor each trait, grid cell and scenario as the percentage deviation of the species-level prediction from the individual-level mean: , $(P_{\text{species}} - \overline{P}_{\text{ind}})/|\overline{P}_{\text{ind}}| \times 100$. Positive values indicate that , where positive values indicate overestimation by the species-level modeloverestimates relative to the individual-level mean, negative values indicate underestimation, and values near zero indicate close agreement. Relative bias can become vary large in grid cells . Because relative bias is undefined or unstable where the individual-level mean is zero or close to zero. Thus, we summarised distributions of relative bias it across the species' range using by its median and by the 95th percentile of the absolute relative biasits absolute value (i.e., the magnitude below which 95% of grid-cell estimates fall).

# 3 Results

## 3.1 Individual physiological variation

Individual DEB parameters varied substantially across the 37 individuals, with coefficients of variation (CV) spanning from 1.7% ($z$) to 8% ($[p_M]$; **Fig. S2.1**). Among the four free individually estimated parameters, the volume-specific somatic maintenance cost $[p_M]$ was the most variable (CV = 8%), followed by energy conductance $\dot{v}$ (CV = 5.6%). The zoom factors ($z$: CV = 1.7%; $z_m$: CV = 1.9%) were less variable. The distribution of each parameter across the 37 individuals followed approximately a normal distribution appeared approximately normal around the species-level estimate (**Fig. S2.2**). In addition, comparisons between observed and DEB-predicted growth trajectories (**Fig. S2.3**) and metabolic rates (**Fig. S2.4**) confirm that both species-level and individual-level parameterisations reproduced the empirical data accurately. Example 20-year DEB simulations for the species-level (female and male) and selected individuals are shown in **Figs. S2.5–S2.14**.

## 3.2 Physiological averaging generates Jensen bias

Under the current climate, forecasts predictions generated from the species-level DEB parameterisation consistently differed differed systematically from the mean of the individual-level forecasts, demonstrating systematic Jensen bias across most ecological performance metrics. The predictions for most performance metrics, although the direction and magnitude of these differences varied among metrics. For females, the species-level parameterisation underestimated most performance metrics relative to the individual-level mean for females (**Fig. 2**). The largest biases occurred for , including basking time ($|J| = 419\,\text{h}$), foraging time ($|J| = 143\,\text{h}$)and , lifetime fecundity ($|J| = 63\,\text{eggs}$). Food intake and maximum body weight were also underestimated, by , food intake ($|J| = 3.7\,\text{g}$and ) and maximum body weight ($|J| = 0.2\,\text{g}$, respectively; here and below, reported values are the absolute value of the mean $J$ across grid cells). Relative to the individual-level mean, bias was largest for $R_0$ (normalised $|J| = 13.3\%$), lifetime fecundity (10.1%) and maximum body weight (5.3%), and smallest for body temperatures ($\leq 0.2\%$). Only maximum and

minimum body temperature showed a consistent positive bias, but with negligible magnitudes (~~< 0.002~~mean $|J| < 0.004$ °C). The pattern for males was largely opposite: the species-level parameterisation overestimated most metrics, ~~with the largest biases for~~ including reproductive energy ($|J| = 2315$ J), basking time ($|J| = 286$ h) and foraging time ($|J| = 398$ h) (**Fig. S3.1**). In relative terms, male bias was largest for reproductive energy (normalised $|J| = 9.1\%$), $R_0$ (7.4%) and food intake (6.9%).

The direction of Jensen bias differed among ecological performance metrics. Underestimation dominated across most of the geographic range for body size, reserve, food, fecundity and basking metrics ($J < 0$ in 63–98% of cells), whereas maximum and minimum body temperature showed the opposite pattern, with predominantly positive bias ($J > 0$ in ~~83–65~~65–83% of cells). Foraging time showed a more balanced distribution (52% positive, 46% negative and 2% with no bias). For males, the pattern was reversed: overestimation dominated for body size, reserve, food, reproductive energy and basking metrics ($J > 0$ in ~~96–57~~57–96% of cells), whereas maximum and minimum body temperature showed predominantly negative bias ($J < 0$ in 84–93% of cells).

The magnitude of physiological heterogeneity ~~itself~~ also varied geographically ~~. The~~ (**Figs. S3.2, S3.3**). Compared with central latitudes (40°–60° N), the median coefficient of variation among individual-level predictions ~~(**Figs. S3.2, S3.3**) was highest toward the northernmost and southernmost latitudes included in the modeling.~~ was higher towards the southern (<40° N) or northern (>60° N) margins of the modelled area, but the strength of this pattern differed markedly among metrics and between sexes. In females,

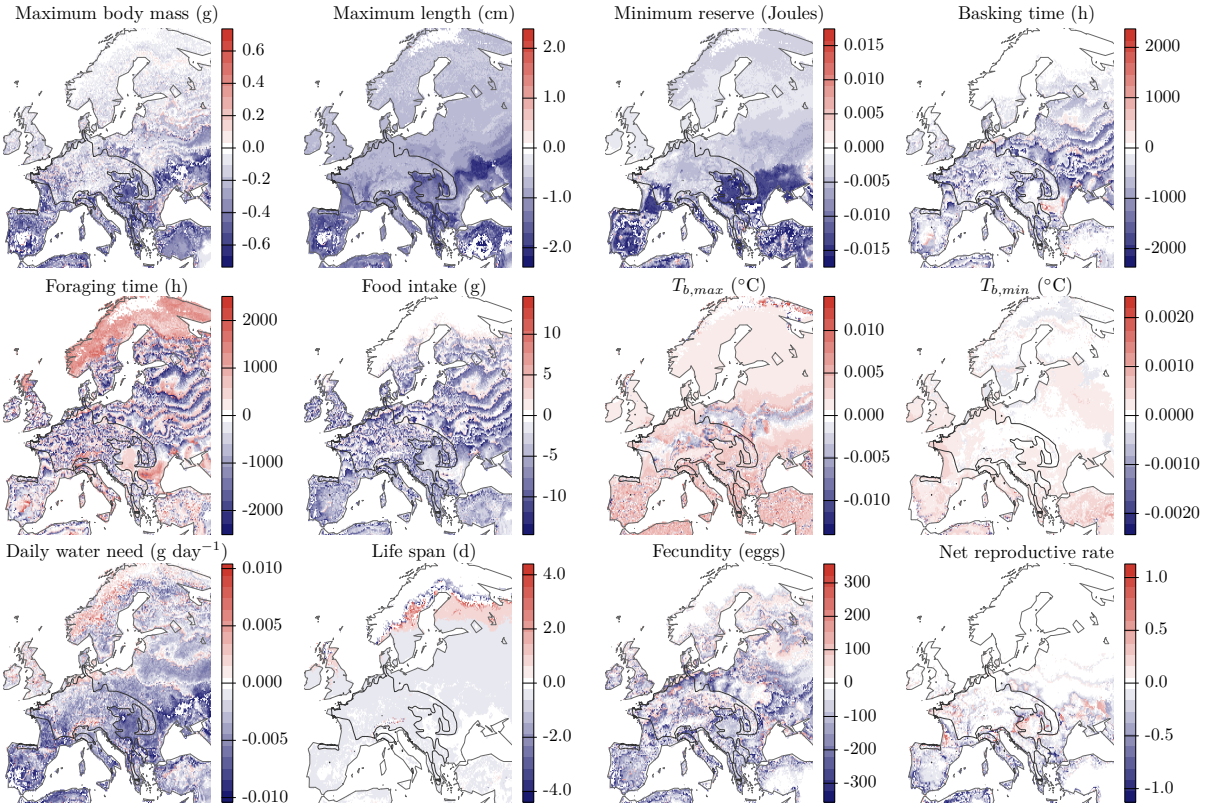


**Figure 2:** Jensen's bias ($J$) between species-level and mean individual-level prediction surfaces for females under the current climate. ~~Values~~ Negative values indicate underprediction by the species-level parameterisation, whereas positive values indicate overprediction; values close to zero indicate close agreement~~; departures from zero indicate systematic differences in the spatial pattern~~. The colour scale is symmetric and centred at zero. $T_b$, body temperature.

the median CV at the southern margin was 3.3-fold higher than at central latitudes for lifetime fecundity, 3.1-fold for minimum reserve and 3.4-fold for maximum body length. In males, the largest increases occurred at the northern margin for $R_0$ (4.3-fold) and reproductive energy (2.2-fold), and at the southern margin for minimum reserve (3.2-fold). In contrast, lifespan, basking time, foraging time and body temperatures differed by at most 1.5-fold between margins and central latitudes in either sex. Thus, the geographic pattern was not universal, but was mainly driven by reproduction- and reserve-related metrics and, in females, by growth-related metrics.

## 3.3 Climate warming reshapes Jensen bias

~~Climate warming significantly altered the magnitude of Jensen bias for all 12 ecological performance metrics (paired Wilcoxon signed-rank tests, all $p < 0.05$; **Fig. 3**). However, warming did not produce a uniform response. For males, the same pattern was observed: warming significantly altered~~ Climate warming altered the magnitude of Jensen bias for all ~~12 metrics(paired Wilcoxon signed-rank tests, all $p < 0.05$), amplifying it for some metrics while attenuating it for others (~~performance metrics, but the direction and magnitude of these changes varied among metrics and between sexes (~~**Fig**~~**Figs. 3, S3.4**).

Under +4 °C warming (**Fig. 3**), the absolute Jensen bias $|J|$ changed ~~significantly~~ for all 12 metrics: the strongest relative amplification occurred for minimum body temperature (rel. $\Delta J = +568\%$; 72% of cells) and maximum body temperature (rel. $\Delta J = +182\%$; 73% of cells), followed by $R_0$ (rel. $\Delta J = +138\%$); in contrast, lifespan ($-15\%$) and minimum

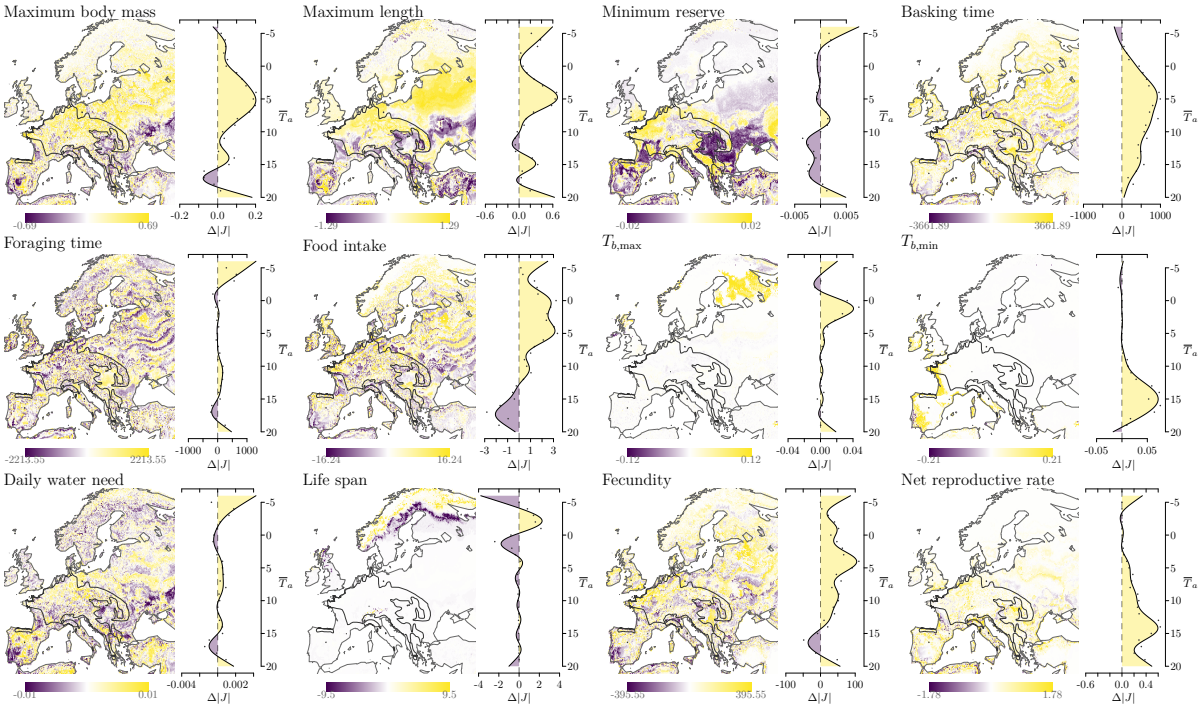


**Figure 3:** Jensen's bias amplification under +4 °C warming. Each map displays the change in absolute Jensen ~~dissimilarity~~ bias ($\Delta|J|$, left, with colour bar) for each performance metric. The curve (descriptive GAM smoother~~and 95 % confidence band~~) at the right of each map represents how $\Delta|J|$ changes along the mean annual temperature ($\overline{T}_a$) gradient. ~~*Yellow* shading~~ Positive values (~~positive $\Delta|J|$~~*yellow*) ~~indicates amplification of the species-averaging bias under~~ indicate that warming ~~, while *purple* shading~~ increases the disagreement between species-level and mean individual-level predictions (amplification), whereas negative ~~$\Delta|J|$~~values (*purple*) ~~indicates~~ indicate reduced disagreement (attenuation).

reserve (−6%) were attenuated. Note, however, that the large relative changes in body temperature bias arise from very small absolute biases: mean $|J|$ increased from 0.0021 to 0.014 °C for minimum and from 0.0035 to 0.0099 °C for maximum body temperature. For males, the strongest amplification occurred for minimum body temperature (rel. $\Delta J = +510\%$; 60% of cells) and maximum body temperature (rel. $\Delta J = +215\%$; 62% of cells), whereas lifespan (−1%) was attenuated.

Warming also reversed the direction of Jensen bias in some regions, causing species-level projections to switch between overestimating and underestimating mean individual performance. The sign of $J$ changed in ~~2.2–49.46~~2–49% of cells between current and +4 °C scenarios, with highest rates for minimum body temperature (49%), foraging time (47%) and $R_0$ (44%). The spatial maps revealed geographically structured changes in Jensen bias, which were reflected in systematic relationships between $\Delta|J|$ and mean annual temperature. However, both the shape and direction of these temperature relationships differed among ecological performance metrics (**Fig. 3**). For males, the sign of $J$ changed in ~~4.61–46.18~~5–46% of cells, with highest rates for basking time (46%), foraging time (43%) and minimum reserve (31%).

Even under +2 °C warming, the direction of change was already discernible for most metrics. 11 metrics showed increased $|J|$, with relative changes ranging from +4% (daily water need) to +152% (minimum body temperature), while 1 metric was attenuated: minimum reserve (−5%). 1 metric (lifespan) switched from amplification at +2 °C to attenuation at +4 °C, suggesting a nonlinear threshold above which warming reverses the species-averaging bias for this trait. For males, 11 metrics showed increased $|J|$, with relative changes ranging from +10% (maximum body length) to +126% (minimum body temperature), while 1 metric was attenuated: minimum reserve (−1%). 1 metric (lifespan) switched from amplification at +2 °C to attenuation at +4 °C.

## 3.4 Consequences for ecological forecasts

~~Jensen bias produced trait-specific differences between forecasts generated from species-level and individual-level physiological parameterisations under both current and future climates~~ The consequences of physiological averaging for ecological forecasts differed markedly among life-history traits and climate scenarios (**Fig. 4**). ~~The magnitude of these differences depended on the ecological trait considered~~For females, the species-level parameterisation underpredicted all three traits in most grid cells, but the magnitude of the discrepancy differed among traits. Lifetime fecundity ~~exhibited~~ showed the largest discrepancies: the species-level model ~~generally underpredicting reproductive output relative to the mean of the individual-level forecasts~~ underpredicted fecundity in 63% of cells under the current climate ~~, although this relationship changed under warming . Lifespan showed a consistent tendency for the species-level model to predict lower values than the individual-level mean in the majority of grid cells across climate scenarios. In contrast, maximum body mass showed comparatively close agreement between species-level~~ (median relative bias −7.1%), and underprediction became more widespread under warming (71% and 76% of cells under +2 °C and ~~individual-level forecasts under all climate scenarios. For males, reproductive energy exhibited the largest discrepancies, with~~ +4 °C, respectively). Lifespan was underpredicted in 76% of cells under the current climate (86% under +4 °C), although the typical relative bias was small (median −1.9%). Maximum body mass was underpredicted in 83% of cells (median relative bias −3.8%). Males showed the opposite direction of bias (**Fig. S3.5**):

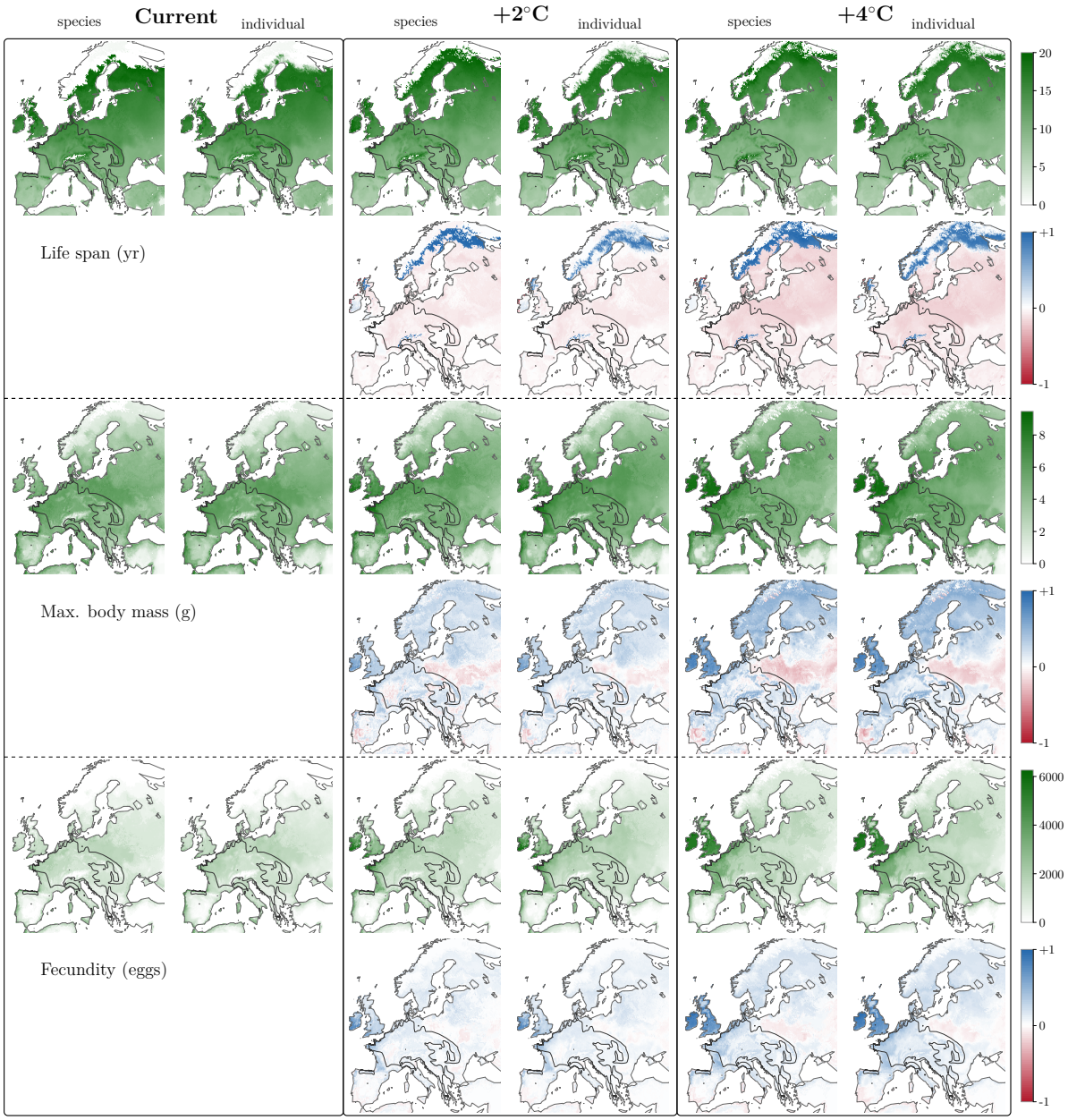


**Figure 4:** ~~Comparison of species-level~~ Species-level and mean individual-level predictions ~~for life span~~and projections of lifespan, maximum body mass and lifetime fecundity under the current ~~,~~climate and +2 °C and +4 °C ~~climate~~ warming scenarios. For each ~~metric~~trait, the top row shows absolute predictions from the ~~raw predicted values for~~ species-level parameterisation and the mean of the individual-level parameterisations under each climate scenario, with all six ~~parameterization–climate combinations~~ maps displayed on a ~~shared~~ common trait-specific scale (*green*). ~~At the~~ The bottom ~~, we show~~ row shows the projected change under +2 °C and +4 °C warming relative to ~~current climates for~~ the ~~warming scenarios~~ corresponding current-climate prediction ($P_{\text{future}} - P_{\text{current}}$)~~on a~~. For visualisation, projected changes were normalized within each trait by the maximum absolute change across all grid cells, parameterization levels, and warming scenarios, yielding a common scale ~~[−1, 1]~~ from -1 to 1. Negative values (~~*blue* indicates decrease, while *red*indicates increase~~) indicate decreases under warming, positive values (*blue*) indicate increases, and values near zero indicate little change.

the species-level model ~~overpredicting reproductive output relative to the individual-level mean~~ overpredicted reproductive energy in 85% of cells under the current climate ~~. This relationship also changed under warming. Lifespan showed a consistent tendency for the species-level model to predict higher values than the individual-level mean across climate~~

scenarios. As for females, (median relative bias +8.1%), increasing to 95% of cells under +4 °C (median +10.9%), and also overpredicted lifespan (84% of cells; median +2.1%) and maximum body mass showed close agreement (**Fig. S3.5**). (95% of cells; median +5.5%). Despite these differences, both parameterisations projected the same direction of change under +4 °C warming in at least 90% of grid cells for all three traits in both sexes.

The relative bias maps (**Figs. S3.6, S3.7**) revealed that, for females, the species-level model underestimated lifespan by up to 19% and fecundity by up to 100% of the individual-level mean, whereas maximum body mass showed relative biases within ±19% for 95% of cells. Under warming, the spatial pattern and magnitude of relative bias shifted, with some regions showing reversed sign. For males, the species-level model overestimated reproductive energy by up to 66% of the individual-level mean, whereas lifespan and maximum body mass showed smaller relative biases.

# 4 Discussion

Mechanistic models have become central to forecasting biological responses to climate change because they explicitly link abiotic environmental conditions to organismal performance through physiology (Fenollosa et al., 2025; Pincebourde & Woods, 2020; Reyes-Puig et al., 2026). Yet, in most cases species are represented by a single average phenotype. Our results show that this assumption introduces systematic bias into mechanistic forecasts: predictions generated from an average physiological phenotype consistently differed from those obtained by propagating empirically measured physiological variation among individuals. Importantly, , and that this bias was not a fixed property of the model, but depended on the performance trait considered, varied geographically across the species' range, and changed under climate warming. These findings do not diminish the value of mechanistic models, which remain among the most powerful approaches for forecasting responses under novel environmental conditions (Fordham et al., 2018; Kearney & Porter, 2009; Maino et al., 2016). Rather, they show that, whenever empirical estimates of individual physiological variation are available, incorporating this information can identify where species-level parameterisations provide robust predictions and where physiological averaging is likely to bias ecological forecasts, consistent with recent calls to increase the biological realism of mechanistic models by explicitly representing within-population heterogeneity (Briscoe et al., 2023; Grimm et al., 2025; Jeltsch et al., 2025).

Our results demonstrate that physiological variation is sufficient to generate systematic forecasting bias. All simulations experienced Because all simulations shared identical microclimatic forcing variables and model structure, differing only in their physiological parameterisation. Consequently, the discrepancies between species-level and individual-level forecasts can only arise from how physiological differences propagate through the nonlinear processes governing growth, maintenance and reproduction (**Fig. 1**). This is precisely the situation in which Jensen's inequality predicts that averaging physiological traits before nonlinear organism–environment interactions is not equivalent to averaging ecological performance afterwards (Bolnick et al., 2011; Ruel & Ayres, 1999). Previous studies have shown analogous amplification arising from temporal environmental variability (Denny, 2017; Paaijmans et al., 2013), while theoretical work predicts that Jensen effects become stronger as the dimensionality of trait variation

increases (Ingram et al., 2018). Our results extend these ideas by showing that variation in only ~~four ecophysiological traits (~~three ecophysiological traits per individual (body size via the sex-specific zoom factor, energy conductance and somatic maintenance; *sensu* Kooijman et al., 2021; Marn and Kooijman, 2022) is sufficient to generate systematic differences in continental-scale mechanistic forecasts, providing empirical evidence that, although inherently difficult to collect (Wolf & McKechnie, 2023), within-population physiological heterogeneity is an important determinant of ecological predictions. These findings do not diminish the value of mechanistic models, which remain among the most powerful approaches for forecasting responses under novel environmental conditions (Fordham et al., 2018; Kearney & Porter, 2009; Maino et al., 2016). Rather, whenever empirical estimates of individual physiological variation are available, incorporating them can identify where species-level parameterisations provide robust predictions and where physiological averaging is likely to bias ecological forecasts, consistent with recent calls to increase the biological realism of mechanistic models by explicitly representing within-population heterogeneity (Briscoe et al., 2023; Grimm et al., 2025; Jeltsch et al., 2025) .

The consequences of physiological heterogeneity were neither uniform among traits nor across environmental gradients. Physiological variation generated systematic, directional bias whose magnitude and, in some cases, even its sign depended on the ecological trait, sex and environmental context (**Figs. 2, S3.1**) (Bolnick et al., 2011; Denny, 2019; Ruel & Ayres, 1999). ~~Under current climate, the species-level parameterisation underestimated basking time, foraging time and lifetime fecundity in females, together with smaller underestimates of food intake and maximum body weight. Males showed the opposite pattern for the same traits, with the largest overestimation for reproductive energy, basking time and foraging time. Relative biasfollowed the same asymmetry: the species-level model underestimated female fecundity and lifespan by up to 100% and 19% of the individual-level mean, respectively, while overestimating male reproductive energy by up to 66%. Maximum body mass, by contrast, was comparatively insensitive to physiological averaging in both sexes~~In both sexes, reproductive output showed the largest relative bias, whereas the direction of bias was largely opposite between females (predominantly underestimation) and males (predominantly overestimation). Foraging time occupied an intermediate position, repeatedly shifting between over- and underestimation across the species' range, indicating that even the direction of Jensen bias is determined by local environmental conditions rather than by the trait itself (Martin & Huey, 2008). Together, these patterns indicate that traits integrating cumulative energy allocation over the life cycle, particularly reproduction, are considerably more sensitive to physiological averaging than traits dominated by structural growth, such as body mass (Nisbet et al., 2012; Palmer et al., 2024). Reproductive output integrates energy acquisition, allocation and maturation history throughout adulthood (Jusup et al., 2017; Kooijman, 2010), allowing relatively small physiological differences to accumulate through time, whereas body mass is constrained by tighter feedbacks between energy acquisition and maintenance (Lika et al., 2019; Nisbet et al., 2012) that might buffer the consequences of individual variation. The opposite direction of bias observed in females and males ~~further reflects~~ may partly arise from their contrasting reproductive allocation strategies, with batch reproduction amplifying energetic differences among females and continuous depletion of the reproductive buffer moderating them in males (Muller et al., 2019; Pecquerie et al., 2009).

Physiological heterogeneity also had geographically structured consequences, indicating

that the importance of individual variation depends on where populations occur within environmental space. Variation among the individual-level predictions was ~~consistently~~ generally highest towards the northernmost and southernmost latitudes represented in the simulations, particularly for reproduction- and reserve-related metrics (**Figs. S3.2, S3.3**), showing that comparatively small physiological differences among individuals became increasingly important under environmentally constraining conditions. This result is consistent with theoretical predictions that the ecological consequences of trait variation emerge from the interaction between the amount of variation present and the local geometry of the underlying performance surface, which changes across environmental gradients (Ross et al., 2023). In other words, the same degree of physiological variation can have relatively small ecological consequences in some environments but substantially larger ones in others (Laughlin & Messier, 2015; Li et al., 2021), depending on where populations operate on the performance surface (Polazzo et al., 2024). Although the strongest effects occurred beyond parts of the current realised distribution of *I. alpestris*, they emerged under the most environmentally challenging conditions represented in the simulations, consistent with the expectation that ecological responses become increasingly sensitive as populations approach environmental limits (Holt, 2003; Sexton et al., 2009; Valladares et al., 2014). This pattern suggests that forecasts based on species-average physiology are therefore most likely to diverge from forecasts that explicitly represent individual variation precisely in situations where predicting persistence, range expansion and local extinction is most relevant (DeMarche et al., 2019).

Furthermore, climate warming altered not only ecological performance itself, but also the magnitude and, in some cases, even the direction of the bias introduced by physiological averaging (**Figs. 3, S3.4**). Climate warming ~~significantly~~ changed Jensen bias for all twelve ecological performance metrics in both sexes, but these changes were strongly trait dependent rather than uniform~~. Under warming, Jensen bias increased approximately 5.7-fold for minimum body temperature, 1.8-fold for maximum body temperature and 1.4-fold for~~ : in both sexes, warming amplified the bias in net reproductive rate, ~~whereas it declined for lifespan and~~ the most ecologically relevant amplification, whereas it attenuated the bias in lifespan (and, in females, in minimum reserve energy~~. Males showed the same qualitative pattern, with the strongest amplificationfor minimum and maximum body temperature and net reproductive rate, and attenuation restricted to lifespan~~). Body temperatures showed the largest relative amplification, although their biases remained negligible in absolute terms. Warming also reversed the sign of Jensen bias across ~~up to 49%~~ large parts of the species' range for some metrics, showing that whether species-level parameterisations over- or underestimate ecological performance depends on the environmental conditions under which populations are evaluated. ~~Most~~ Moreover, whereas most metrics already showed increased ~~absolute~~ bias under +2 °C~~warming, but~~ , lifespan switched from amplification at +2 °C to attenuation at +4 °C, demonstrating that the influence of physiological heterogeneity on ecological forecasts ~~do~~ does not change monotonically with warming. Instead, warming can shift populations across different regions of nonlinear performance surfaces (Angilletta Jr., 2009; Kingsolver & Woods, 2016; Li et al., 2021), such that further temperature increases weaken rather than strengthen the effects of physiological averaging. These results are consistent with the broader view that biological responses to climate change emerge from nonlinear interactions between organismal traits and environmental conditions, so that warming changes where populations operate on underlying performance surfaces rather than simply scaling biological responses (Buckley et al., 2018, 2023; Cannistra & Buckley, 2024). Our results show that these same

nonlinearities also determine where forecasts based on average physiology provide robust predictions and where they systematically diverge from forecasts that explicitly account for physiological heterogeneity.

Despite the systematic biases introduced by physiological averaging, both physiological parameterisations converged on the same broad biological response of *I. alpestris* to climate warming, projecting the same direction of change in at least 90% of grid cells for lifespan, body mass and reproduction (**Figs. 4, S3.5**). Under both parameterisations, warming reduced projected lifespan across most of the species' distribution, consistent with the faster life histories and higher cumulative metabolic costs expected for ectotherms under warmer conditions (Burraco et al., 2020; Munch & Salinas, 2009). At the same time, both approaches projected reduced body mass and reproductive output towards the north-eastern limit of the species' range, where climatic warming is expected to increase thermal suitability. This combination suggests that improved thermal conditions alone may not translate directly into greater demographic potential if gains in climatic suitability are accompanied by reductions in growth and reproductive performance (Csergő et al., 2017; Schulte, 2015). Thus, incorporating individual physiological variation did not fundamentally alter the biological interpretation of the species' response to climate change, but instead revealed where and by how much forecasts became sensitive to the simplifying assumption of representing populations by a single physiological phenotype. Because mechanistic models explicitly represent organismal physiology, they are particularly well suited to incorporate intraspecific physiological variation, phenotypic plasticity and evolutionary responses directly into ecological predictions (Riddell et al., 2023). Previous studies have propagated observed or hypothetical physiological variation through mechanistic models to investigate trait evolution, adaptive significance and thermal adaptation (e.g., De Jong et al., 1996; Enriquez-Urzelai et al., 2020; Kingsolver and Woods, 2016; Riddell et al., 2022). ~~Our~~ ; our results show that the same framework can also ~~be used to~~ quantify where the robustness of mechanistic forecasts changes across ecological traits, geographic space and future climates. ~~Species-average parameterisations therefore provide a useful first approximation of expected responses, whereas propagating empirically measured physiological variation identifies where those predictions remain reliable and where they become intrinsically sensitive to physiological averaging.~~

As individual-level physiological datasets become increasingly available (Albert et al., 2010; Briscoe et al., 2015; Fyllas et al., 2020; Gardner et al., 2019), propagating trait distributions offers a practical way to quantify ~~biological sources of uncertainty~~ the consequences of biological heterogeneity alongside the increasingly realistic representation of environmental heterogeneity. Realising this potential, however, depends on the availability of empirical data describing physiological variation among individuals. Physiological studies commonly report population or treatment means, often accompanied only by summary measures of dispersion, whereas the individual-level estimates on which those summaries are based are rarely reported (Wolf & McKechnie, 2023). Yet, it is precisely the distribution of individual phenotypes, rather than its central tendency, that determines whether and how strongly forecasts based on average phenotypes are biased. Thus, we strongly encourage empiricists to report individual-level estimates (or at least their full distributions and covariances) alongside population means, ideally in open repositories. Such data would allow us to propagate realistic trait distributions through mechanistic models, to evaluate how biased or robust our predictions are, and to identify the regions within the distribution of an species where within-species phenotypic variation will aid species the most in

coping with climate change.

Several limitations also point to important directions for future work. We only estimated individual variation for ~~four out of the twelve primary DEB parameters~~ three DEB parameters per individual (the sex-specific zoom factor, $\dot{v}$ and $[\dot{p}_M]$). Variation in the remaining ecophysiological traits or in other traits related to morphology, thermal biology and behaviour could further modify both the magnitude and the spatial pattern of Jensen bias. Likewise, all individuals originated from a single population, whereas physiological trait distributions are likely to vary across species' ranges due to local adaptation, developmental history and long-term acclimation (Enriquez-Urzelai et al., 2020; Moran et al., 2016; Sexton et al., 2009). ~~Rather than weakening our conclusions, however, we believe that incorporating~~ Incorporating such among-population variation ~~would likely introduce additional systematic, spatially structured forecasting bias~~ could amplify, attenuate or even reverse the spatial patterns of Jensen bias reported here. Finally, we assumed that physiological differences remained fixed under future climates, whereas phenotypic plasticity and evolutionary responses are expected to reshape trait distributions over the timescales relevant to climate change (Chevin et al., 2010; Kingsolver & Buckley, 2017; Riddell et al., 2023). Extending mechanistic forecasts to account simultaneously for individual variation, geographic differentiation and the dynamic reshaping of trait distributions through plasticity and adaptation therefore represents an important next step towards biologically realistic predictions under environmental change.

# 5 Conclusion

Our results demonstrate that physiological heterogeneity systematically influences mechanistic climate forecasts. Forecasts generated from a single average physiological phenotype consistently differed from those obtained by propagating empirically measured individual variation, with the magnitude and direction of Jensen bias depending on the ecological trait considered, varying geographically across the species' range, and changing under climate warming. At the same time, species-average parameterisations generally recovered the broad qualitative responses of *I. alpestris* to warming, indicating that physiological heterogeneity primarily influences the magnitude and spatial structure of forecasts rather than their overall direction. Together, these findings show that biological heterogeneity should be considered alongside environmental heterogeneity in mechanistic climate forecasting. As environmental datasets and microclimate models become increasingly realistic, incorporating empirically measured physiological variation represents a natural next step towards improving the biological realism of ecological predictions. Representing populations as distributions of physiological phenotypes, rather than single average individuals, offers a practical path towards forecasts that quantify not only expected ecological responses to climate change, but also where those predictions are robust and where they are intrinsically susceptible to bias arising from nonlinear averaging.

# Author Contributions

**Urtzi Enriquez-Urzelai:** writing – original draft, writing – review and editing, conceptualization, formal analysis, methodology, visualization, software. **Gioele Pascucci:** writing – review and editing. **Lumír Gvoždík:** writing – review and editing, conceptualization, funding acquisition.

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# Conflicts of Interest

The authors declare no conflicts of interest.

# Data Availability Statement

All the data and code to reproduce this study are freely available at GitHub (https://github.com/urtzienriquez/individuals_DEB).

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# **Appendix 1** *for:* Beyond the average phenotype: individual variation alters mechanistic forecasts of climate responses

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# Supplementary Materials and Methods

## 1 The standard DEB model

Dynamic Energy Budget (DEB) theory (Kooijman, 2010; Sousa et al., 2008) describes the acquisition, storage, and allocation of energy and mass throughout the life cycle. The standard DEB model (Marques et al., 2018) represents an organism by three state variables: structural volume $V$, reserve energy $E$, and cumulative energy invested in maturation $E_H$ (plus a reproduction buffer $E_R$ in adults). Energy enters the organism as food, is assimilated into reserve at a rate proportional to surface area, and is subsequently mobilised (catabolised) from the reserve to fuel five downstream processes: somatic maintenance, growth in structure, maturity maintenance, maturation (in juveniles), and reproduction (in adults). Reproduction energy accumulates in a buffer and is periodically converted into eggs. Life-history transitions occur at species-specific maturity thresholds: e.g., birth at $E_H^b$ and sexual maturity at $E_H^p$ (Kooijman, 2010; Marques et al., 2018; Oliveira et al., 2026).

The behaviour of the standard model is governed by 12 primary parameters (Kooijman, 2010). The maximum surface-specific assimilation rate $\{\dot{p}_{Am}\}$ controls the rate of energy acquisition and, through covariation rules, scales with maximum body size via the dimensionless zoom factor $z$. A larger $z$ implies a larger body, a proportionally higher assimilation capacity, and a greater energetic investment at life-history transitions (Sousa et al., 2008). Once assimilated, reserves are mobilised at a rate set by the energy conductance $\dot{v}$ and following the $\kappa$-rule: a fixed fraction $\kappa$ of mobilised reserve is allocated to somatic maintenance, whose cost per unit structural volume is $[\dot{p}_M]$, and to growth, at a specific cost of $[E_G]$ per unit of new structure; the remainder $1-\kappa$ supports maturity maintenance (at a rate set by $\dot{k}_J$) and maturation in juveniles or reproduction in adults, with eggs produced at an efficiency $\kappa_R$. The rates of feeding and assimilation are further determined by the maximum specific searching rate $\{\dot{F}_m\}$, the assimilation efficiency $\kappa_X$, and the surface-specific maintenance cost $\{\dot{p}_T\}$, which is assumed to be zero for ectotherms. The cumulative energetic investments at which organisms reach birth and sexual maturity are defined by the thresholds $E_H^b$ and $E_H^p$. Auxiliary parameters include the shape coefficient $\delta_M$, which allows to convert shape-specific length measures to lengths relevant in the DEB context, and the acceleration of ageing $\ddot{h}_a$. Sex-specific parameters can also be used to account for between-sex differences, for instance in male puberty ($E_{H,m}^p$) and body-size dimorphism ($z_m$), and, in species with a larval stage, the standard model can be extended with an additional maturity threshold at metamorphosis, $E_H^j$. Because each parameter quantifies the rate or efficiency of a

specific physiological process, the standard DEB model provides a coherent framework for interpreting individual variation in terms of energetic traits with distinct ecological meaning (Kooijman et al., 2021).

## 2 DEB parameter estimation

We fitted the standard DEB model for *I. alpestris* by inverse modelling, using the covariation method (Lika et al., 2011; Marques et al., 2018) as implemented in the freely available AmPtool and DEBtool_M packages for MATLAB (The MathWorks Inc.). Following standard practice in this method, each data set was assigned the temperature at which it was obtained, and rates were corrected for temperature with the Arrhenius relationship. Model goodness-of-fit was assessed with the mean relative error (MRE) and the symmetric mean squared error (SMSE) (Marques et al., 2018).

For the species-level parameterization, the model was fitted simultaneously to the species-level life-history information and the complete set of longitudinal growth and metabolic observations from all 37 individuals. Individual observations were retained as separate data points rather than averaged across individuals or ages, and male and female observations were fitted jointly within the same parameter-estimation procedure, while allowing sex-specific parameters. Of the 12 primary parameters, we estimated seven: the maximum surface-specific assimilation rate $\{\dot{p}_{Am}\}$, via the zoom factor $z$; the energy conductance $\dot{v}$; the allocation fraction $\kappa$; the volume-specific somatic maintenance $[\dot{p}_M]$; the specific cost of structure $[E_G]$; and the maturity thresholds at birth $E_H^b$ and puberty $E_H^p$. The remaining five ($\{\dot{F}_m\}$, $\kappa_X$, $\kappa_R$, $\{\dot{p}_T\}$ and $\dot{k}_J$) were fixed to standard values from the DEB literature (Kearney et al., 2013; Kooijman, 2010). We also estimated the maturity threshold at metamorphosis $E_H^j$, which marks the timing and size of metamorphosis without changing the physiology of the model (i.e., no metabolic acceleration was assumed); the male puberty threshold $E_{H,m}^p$ and the male zoom factor $z_m$, which account for sexual size dimorphism; the shape coefficient $\delta_M$; and the Weibull ageing acceleration $\ddot{h}_a$. The Arrhenius temperature $T_A$ and all chemical parameters were fixed at standard values. Together, these estimated parameters define a model representation of the species' energetic phenotype, with each parameter controlling a distinct aspect of energy acquisition, allocation, or expenditure. Model parameters are given in **Table S2.1**, and the full parameter set and goodness of fit can be found at https://www.bio.vu.nl/thb/deb/deblab/add_my_pet/entries_web/Ichthyosaura_alpestris/Ichthyosaura_alpestris_res.html.

Individual-level parameters were obtained by re-estimating three parameters independently for each individual (the sex-specific zoom factor, $z$ for females and $z_m$ for males; the energy conductance $\dot{v}$; and the volume-specific somatic maintenance $[\dot{p}_M]$), while all remaining parameters, including $\kappa$, $[E_G]$ and the maturity thresholds, were held at their species-level values. Each estimation used the individual's own longitudinal growth and metabolic trajectories as univariate data, with species-level estimates serving as starting values, and each individual was assigned the sex-appropriate puberty threshold ($E_H^p$ for females, $E_{H,m}^p$ for males).

## 3 Microclimate and ectotherm simulations

We reconstructed microclimatic conditions using NicheMapR's microclimate model, as implemented in the `micro_global()` function (Kearney & Porter, 2017; Kearney et al., 2014), which uses monthly climatological means as forcing environmental variables. The model uses monthly climatological data (maximum and minimum air temperature, precipitation, cloud cover, relative humidity, wind speed and number of wet days) from the global 10 arc-minute climatology of New et al. (2002), together with elevation, slope, aspect and soil properties, to estimate hourly aboveground air temperature, solar and infrared radiation, wind speed and relative humidity at the animal's height, as well as soil temperature and moisture profiles at ten depths (Kearney & Porter, 2017; Kearney et al., 2014). Microclimate simulations were run for 22 years with the moisture and snow routines enabled (Kearney, 2020; Kearney & Maino, 2018). The first two years were excluded as spin-up to ensure stabilized snow and soil moisture dynamics before organismal simulations, resulting in simulations of up to 20 years (i.e., the maximum known lifespan of the species). We ran the microclimate simulations in two shading conditions: fully unshaded (0% shade) and deep shade (90% shade).

To estimate *I. alpestris* performance and simulate its life cycle with coupled biophysical and DEB models, we used the `ectotherm()` function of NicheMapR (Kearney & Porter, 2020). We parameterised the ectotherm model for a terrestrial alpine newt, with 90% of the skin surface acting as a free water exchanger to account for evaporative water loss (Enriquez-Urzelai & Gvoždík, 2024). Food availability was held constant throughout the simulations, so that the amount of energy acquired depended on the time available for foraging. Individuals were assumed to be nocturnal and crepuscular, but juveniles were also allowed diurnal activity, reflecting the need to acquire energy reserves before their first winter (Enriquez-Urzelai et al., 2020). Animals retreated underground during daylight hours (in the case of adults), and whenever body temperature fell outside the voluntary thermal limits (i.e., the lower and upper body temperatures within which an individual is assumed to remain behaviorally active; $VT_{min} = 4\,°C$, $VT_{max} = 20\,°C$), or relative humidity aboveground was lower than 65%. This relative humidity value corresponds well with the lowest relative humidity selected by amphibians at low temperatures (Bonnefond et al., 2020; Giacometti & Tattersall, 2025). When burrowed, animals selected a depth (from 2.5 cm up to 2 m) that prevented them from reaching critical thermal limits (i.e., temperatures at which physiological performance fails; $CT_{min} = -2\,°C$ and $CT_{max} = 36\,°C$) (Enriquez-Urzelai & Gvoždík, 2024). The preferred body temperature was set to 17.3 °C (Gvoždík & Kristín, 2017). Although amphibians are not as accurate thermoregulators as non-avian reptiles, we allowed individuals to behaviourally select shaded microhabitats by adjusting the amount of shade needed (from 0% to 90% of shade) to stay within their preferred temperature range (Enriquez-Urzelai et al., 2019).

In the case of females (both at the species and individual level), we simulated reproduction using the batch reproductive model (Pecquerie et al., 2009): females stored energy in the reproductive buffer until it was enough to produce a clutch (200–600 eggs) during the breeding season. In males, we simulated continuous depletion of the reproductive buffer during the breeding season (representing energy invested in sperm, courtship, etc.), so no energy allocated to reproduction was stored. The breeding season occurred between the spring equinox and the summer solstice. We ran the simulations from the juvenile stage (starting on 15 August, when juveniles are present in the field), for the total length of the microclimate simulation, until the individual died.

# 4 Jensen bias and variability metrics

Positive values of Jensen bias ($J = P_{\text{species}} - \overline{P}_{\text{ind}}$) indicate that the species-level parameterisation overpredicts performance relative to the mean of individuals, whereas negative values indicate underprediction. To avoid interpreting numerical noise as bias, we considered grid cells with $|J| \leq 10^{-10}$ as showing no bias, and, when quantifying the proportion of grid cells in which absolute Jensen bias increased, decreased or remained unchanged under warming, we considered changes with $|\Delta|J|| \leq 10^{-10}$ as unchanged. To compare the magnitude of bias among metrics measured in different units, we computed the normalised absolute Jensen bias for each metric and sex under the current climate, as the mean $|J|$ across grid cells divided by the mean $|\overline{P}_{\text{ind}}|$ across grid cells ($\times 100$).

The coefficient of variation among individual-level predictions was computed for each sex (18 females and 19 males), metric and grid cell as $\text{CV} = SD/(|\overline{P}_{\text{ind}}| + \varepsilon)$, where the small constant $\varepsilon = 10^{-10}$ avoids division by zero. To summarise its geographic pattern, we grouped grid cells into southern (<40° N), central (40°–60° N) and northern (>60° N) latitudinal bands and compared the median CV of each marginal band with that of the central band.

# 5 Jensen bias along the temperature gradient

Mean annual air temperature ($\overline{T}_a$) was calculated for each grid cell from the global 10 arc-minute climatology of New et al. (2002) by averaging the monthly means of the daily maximum and minimum air temperatures. The change in absolute Jensen bias under +4 °C warming ($\Delta|J|$) was rasterized at approximately 0.16° resolution for each of the 12 performance metrics and summarized as binned means (±1 SE) in 1 °C $\overline{T}_a$ intervals, retaining only bins containing at least 10 grid cells. To describe the overall trend along the $\overline{T}_a$ gradient, we fitted generalised additive models (GAMs) with the `mgcv` R package (Wood, 2025) to the cell-level $\Delta|J|$ values, separately for each metric and sex. Models assumed a Gaussian error distribution with identity link and included a thin-plate regression spline smooth of $\overline{T}_a$ (basis dimension $k = 12$), with smoothing parameters estimated by restricted maximum likelihood (REML) and an additional shrinkage penalty (`select = TRUE`). Predictions were evaluated at 0.1 °C intervals along the $\overline{T}_a$ gradient. Because model outputs are deterministic and neighbouring grid cells are spatially autocorrelated, we used GAMs as descriptive smoothers rather than for statistical inference, and we did not report confidence intervals.

# 6 Comparison of species- and individual-level forecasts

For each of the three life-history traits (lifespan, maximum body mass and lifetime fecundity, or reproductive energy for males), grid cell and climate scenario, we extracted the species-level prediction ($P_{\text{species}}$) and the corresponding mean individual-level prediction ($\overline{P}_{\text{ind}}$). Comparisons included both the absolute predictions under the current climate and the projected changes under warming, expressed relative to the current climate ($P_{\text{future}} - P_{\text{current}}$) for each physiological parameterisation. For visualisation, we normalized the projected changes to range between −1 and 1 by dividing each grid cell's value by the maximum absolute change within each trait. We then calculated, for each

trait, sex and scenario, the percentage of grid cells in which the species-level model under- or overpredicted the individual-level mean, and the percentage of grid cells in which both parameterisations projected the same direction of change under warming.

We also computed the relative bias for each trait, grid cell and scenario as the percentage deviation of the species-level prediction from the individual-level mean, $(P_{\text{species}} - \overline{P}_{\text{ind}})/|\overline{P}_{\text{ind}}| \times 100$, so that positive values indicate overestimation by the species-level model, negative values underestimation, and values near zero close agreement. Relative bias is undefined in grid cells where the individual-level mean is zero, and these cells were excluded, and it can become very large where the individual-level mean is close to zero. To limit the influence of such near-zero denominators, we summarised the distributions of relative bias across the species' range using the median relative bias and the 95th percentile of the absolute relative bias (i.e., the magnitude of relative bias below which 95% of grid-cell estimates fall).

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# **Appendix 2** *for:* Beyond the average phenotype: individual variation alters mechanistic forecasts of climate responses

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# DEB model parameters and example simulations

## 1 DEB core parameter analysis across individuals

We estimated Dynamic Energy Budget (DEB) model parameters for 37 *I. alpestris* individuals using the standard DEB model (std) (Kooijman, 2010). The standard DEB model is governed by 12 primary parameters that jointly determine the rates of assimilation, mobilisation, maintenance, growth, and reproduction (Marques et al., 2018). At the species level, seven of the 12 primary parameters were freed for estimation; the remaining five were held at their literature values. At the individual level, four parameters were allowed to vary: the zoom factors $z$ (female) and $z_m$ (male), the energy conductance $\dot{v}$, and the volume-specific somatic maintenance rate $[\dot{p}_M]$. All remaining parameters were held at their species-level values.

The zoom factor $z$ scales maximum body size ($L_m = z \cdot L_{\text{ref}}$, with $L_{\text{ref}} = 1$ cm) and, through the physical covariation rules (Kooijman, 2010), simultaneously determines the maximum surface-specific assimilation rate $\{\dot{p}_{Am}\}$ and the maturity thresholds at birth ($E_H^b$) and puberty ($E_H^p$). Because $z$ (or $z_m$) and $[\dot{p}_M]$ vary among individuals while $\kappa$ is fixed, $\{\dot{p}_{Am}\}$ is not a free parameter but is derived from the individual estimates of $z$ and $[\dot{p}_M]$:

$$\{\dot{p}_{Am}\} = \frac{z \cdot L_{\text{ref}} \cdot [p_M]}{\kappa} \qquad \text{and} \qquad \{\dot{p}_{Am,m}\} = \frac{z_m \cdot L_{\text{ref}} \cdot [p_M]}{\kappa} \tag{1}$$

The complete set of parameter values is presented in ~~Table~~ S2.1**Table S2.1**.

**Table S2.1:** DEB parameter values for *I. alpestris*: species-level estimates and individual ranges (min–max) across the 37 individuals.

| Parameter | Symbol | Units | Species | Individuals | |
|---|---|---|---|---|---|
| | | | | Min | Max |
| **Primary parameters** | | | | | |
| max spec searching rate | $\{\dot{F}_m\}$ | $\mathrm{l\,d^{-1}\,cm^{-2}}$ | 6.5 | – | – |
| digestion efficiency of food to reserve | $\kappa_X$ | – | 0.8 | – | – |
| max surface-specific assimilation rate | $\{\dot{p}_{Am}\}$ | $\mathrm{J\,d^{-1}\,cm^{-2}}$ | 265.7 | 217.6 | 303.3 |
| energy conductance | $\dot{v}$ | $\mathrm{cm\,d^{-1}}$ | 0.01914 | 0.01716 | 0.02134 |
| allocation fraction to soma | $\kappa$ | – | 0.5914 | – | – |
| reproduction efficiency | $\kappa_R$ | – | 0.95 | – | – |
| vol-spec somatic maint | $[\dot{p}_M]$ | $\mathrm{J\,d^{-1}\,cm^{-3}}$ | 138.4 | 107.4 | 162.8 |
| surf-spec somatic maint | $\{\dot{p}_T\}$ | $\mathrm{J\,d^{-1}\,cm^{-2}}$ | 0 | – | – |
| maturity maint rate coefficient | $\dot{k}_J$ | $\mathrm{d^{-1}}$ | 0.002 | – | – |
| spec cost for structure | $[E_G]$ | $\mathrm{J\,cm^{-3}}$ | 7302 | – | – |
| maturity at birth | $E_H^b$ | J | 6.126 | – | – |
| maturity at puberty for females | $E_H^p$ | J | $1.366 \times 10^4$ | – | – |
| **Extended parameters** | | | | | |
| zoom factor | $z$ | – | 1.135 | 1.102 | 1.198 |
| zoom factor for males | $z_m$ | – | 1.046 | 0.9978 | 1.118 |
| max surface-specific assimilation rate | $\{\dot{p}_{Am,m}\}$ | $\mathrm{J\,d^{-1}\,cm^{-2}}$ | 244.8 | 191.9 | 307.9 |
| maturity at puberty for males | $E_{H,m}^p$ | J | 7474 | – | – |
| maturity at metam | $E_H^j$ | J | 334.2 | – | – |
| shape coefficient | $\delta_M$ | – | 0.1878 | – | – |
| Weibull aging acceleration | $\ddot{h}_a$ | $\mathrm{d^{-2}}$ | $6.475 \times 10^{-9}$ | – | – |

## 1.1 Parameter variability

The coefficient of variation (CV) quantifies relative variability across individuals. ~~Figure S2.1~~ **Figure S2.1** shows the CV values sorted in descending order. Among the four free parameters, $[\dot{p}_M]$ exhibited the largest relative variation across individuals, followed by $\dot{v}$, while $z$ was the most conservative. The mean individual values deviated from the species-level estimate by 0.9% for $z$, -0.9% for $z_m$, 1% for $\dot{v}$, and -0.8% for $[\dot{p}_M]$.

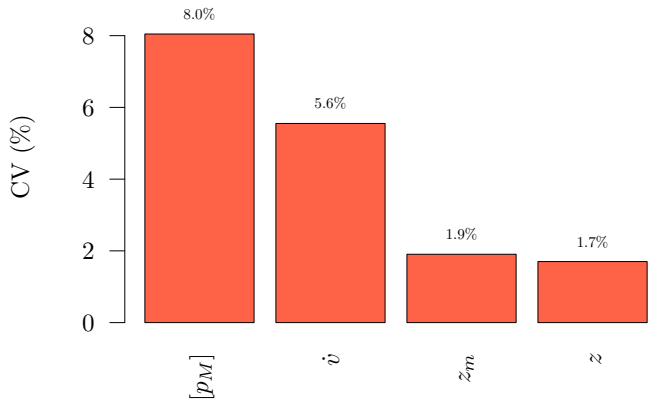


**Figure S2.1:** Coefficient of variation for each varying parameter. $[\dot{p}_M]$ is the most variable parameter across individuals.

~~Figure S2.2~~ **Figure S2.2** shows the distribution of each parameter with the species-level value marked for reference. Most parameters were broadly distributed around the species-level estimate, indicating that the individual-level data span a wider range of physiological strategies than the single species-level value suggests.

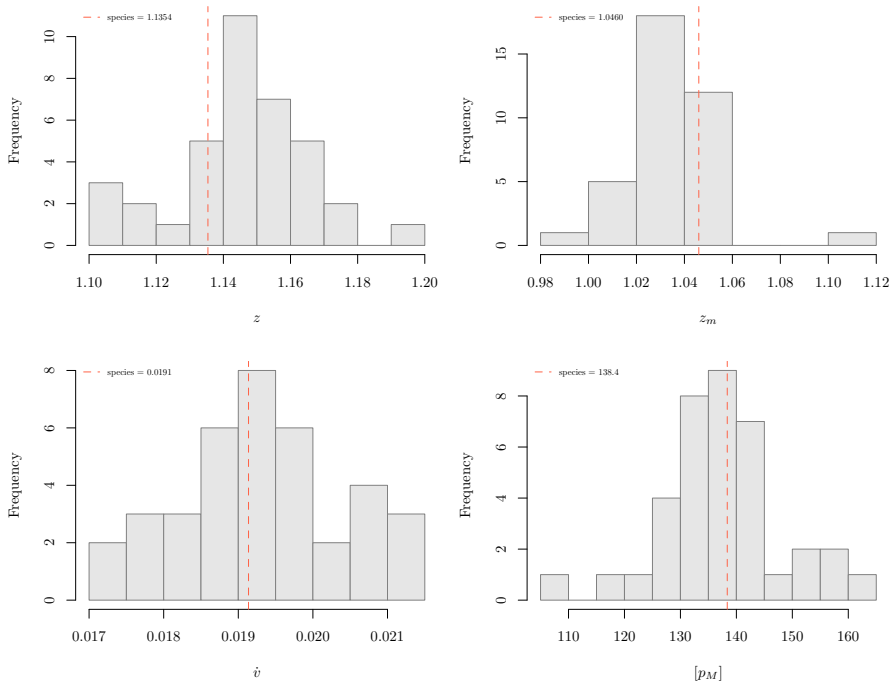


**Figure S2.2:** Distribution of each varying parameter. The dashed line marks the species-level estimate.

# 2 Fit of DEB models to the data

In **Fig. S2.3** and **Fig. S2.4** we show the fit of the model with the final set of estimated parameters at the species level (for each sex in different colors) and for four selected females and males.

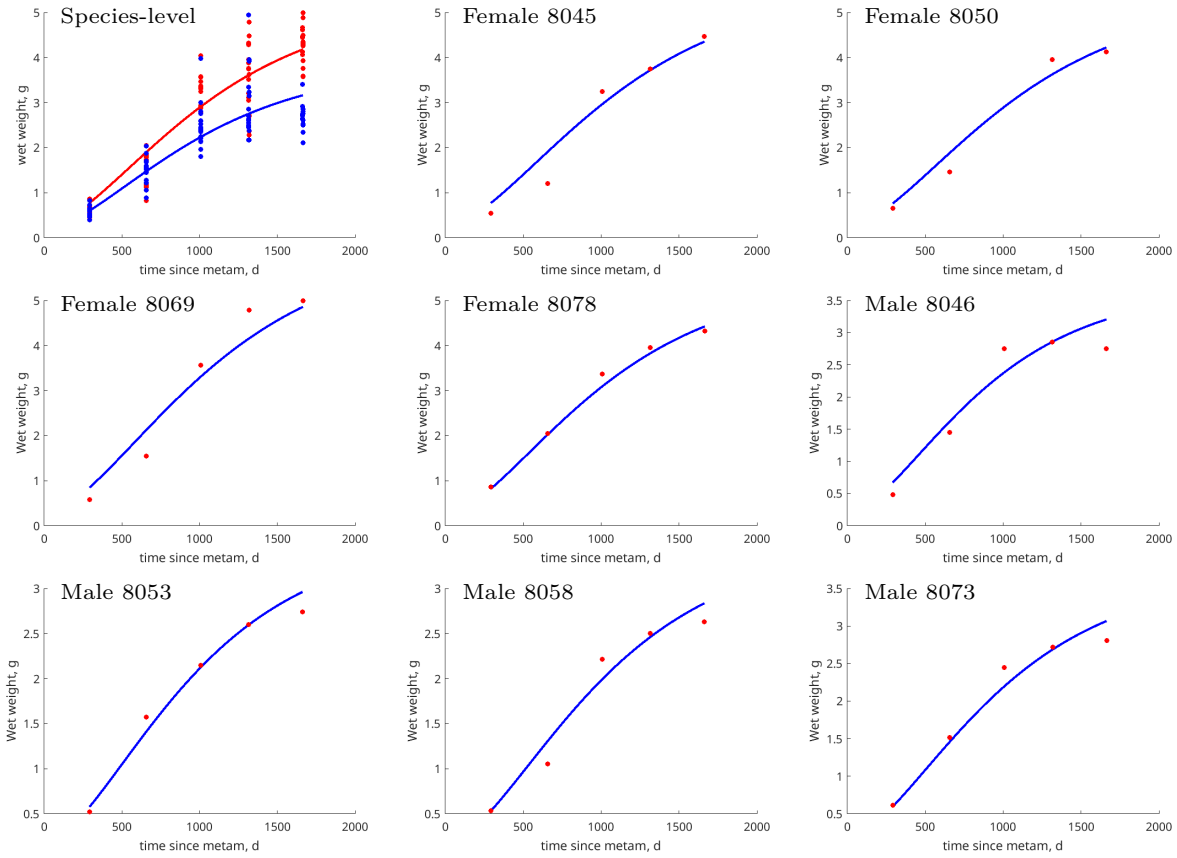


**Figure S2.3:** ~~test figure with the param estimation~~ Model predictions ~~for~~ of growth at the species level (*blue*: males; *red*: females) and eight selected individuals.

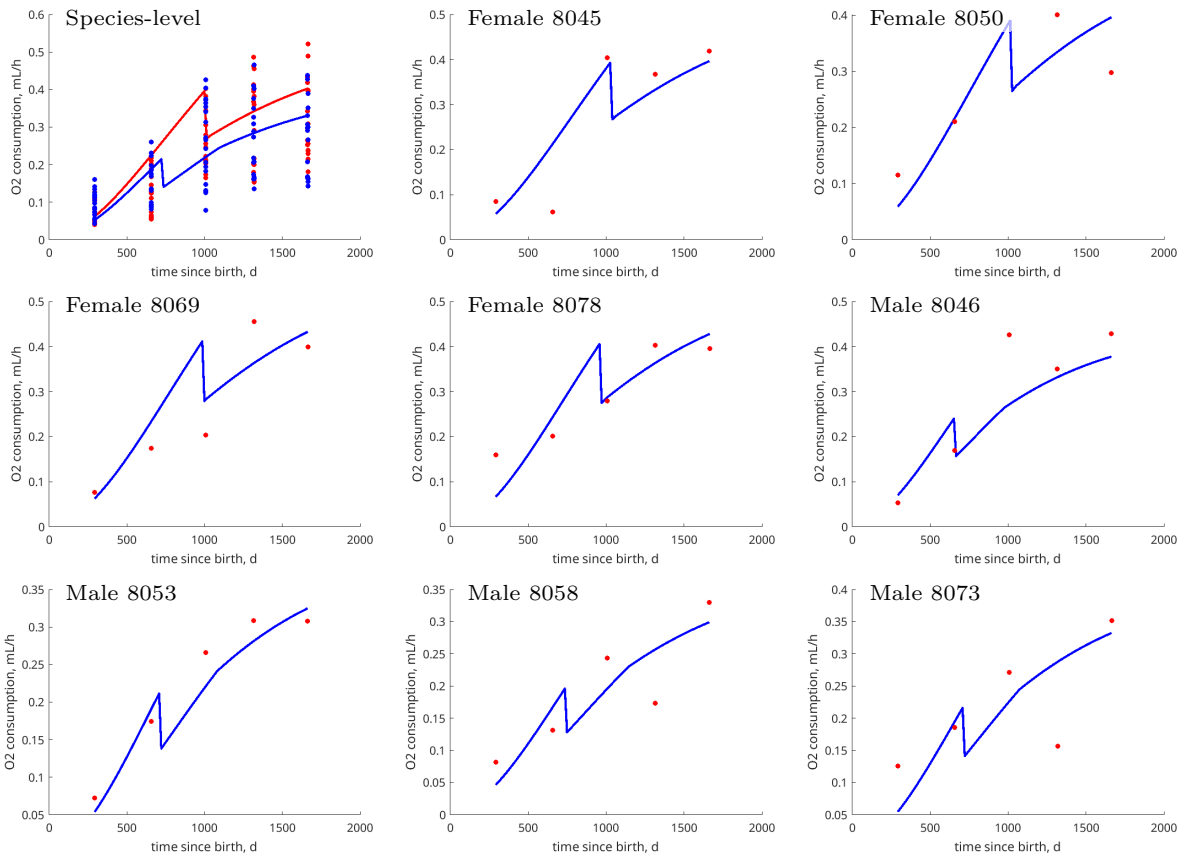


**Figure S2.4:** ~~test figure with the param estimation~~ Model predictions for ~~oxygen consumption~~metabolic rates at the species level (*blue*: males; *red*: females) and eight selected individuals.

# 3 Growth trajectories

Individual 20-year DEB growth simulations at the **(a)** Czech reference site (49.2247° N, 16.0579° E; 600 m a.s.l.) and **(b)** at a comparatively warmer site in Northern Spain (inside the known distribution of the species). A vertical dashed line marks when individuals reached puberty (first hour where maturity $E_H$ exceeds the puberty threshold $E_{Hp}$). We present example simulation for females first (with the species level parameters, and for four selected individuals) and then for males (species level parameters first and for individual-level parameters after). Note that the selected individuals correspond to the individuals shown in **Figs.** S2.3 and S2.4.

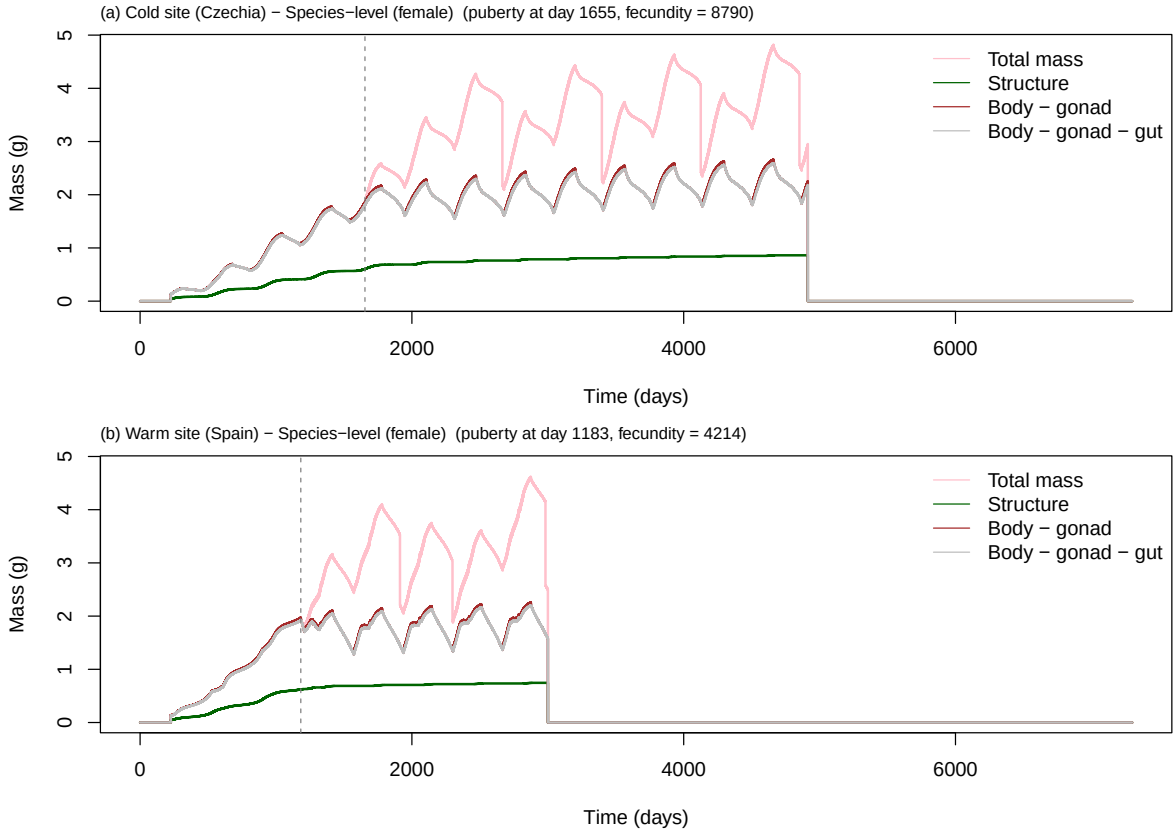


**Figure S2.5:** Growth trajectory for Species-level (female).

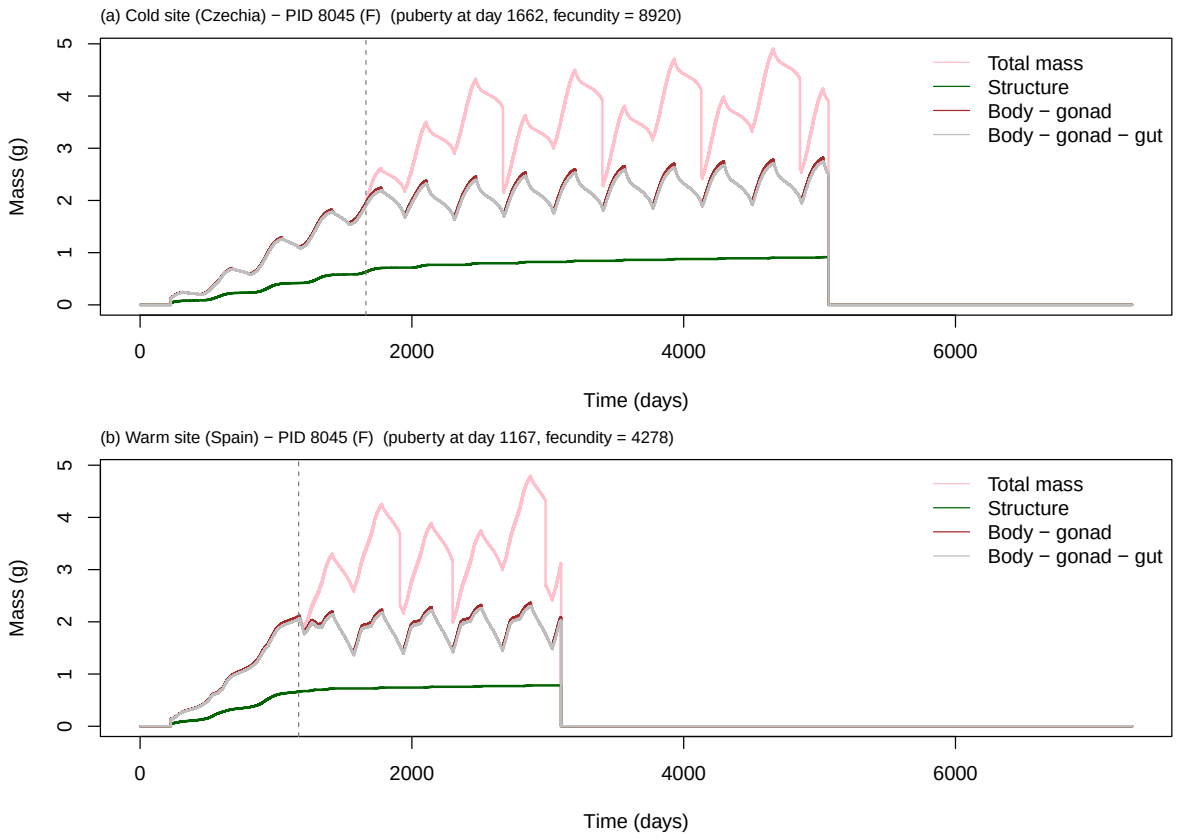


**Figure S2.6:** Growth trajectory for PID 8045 (female).

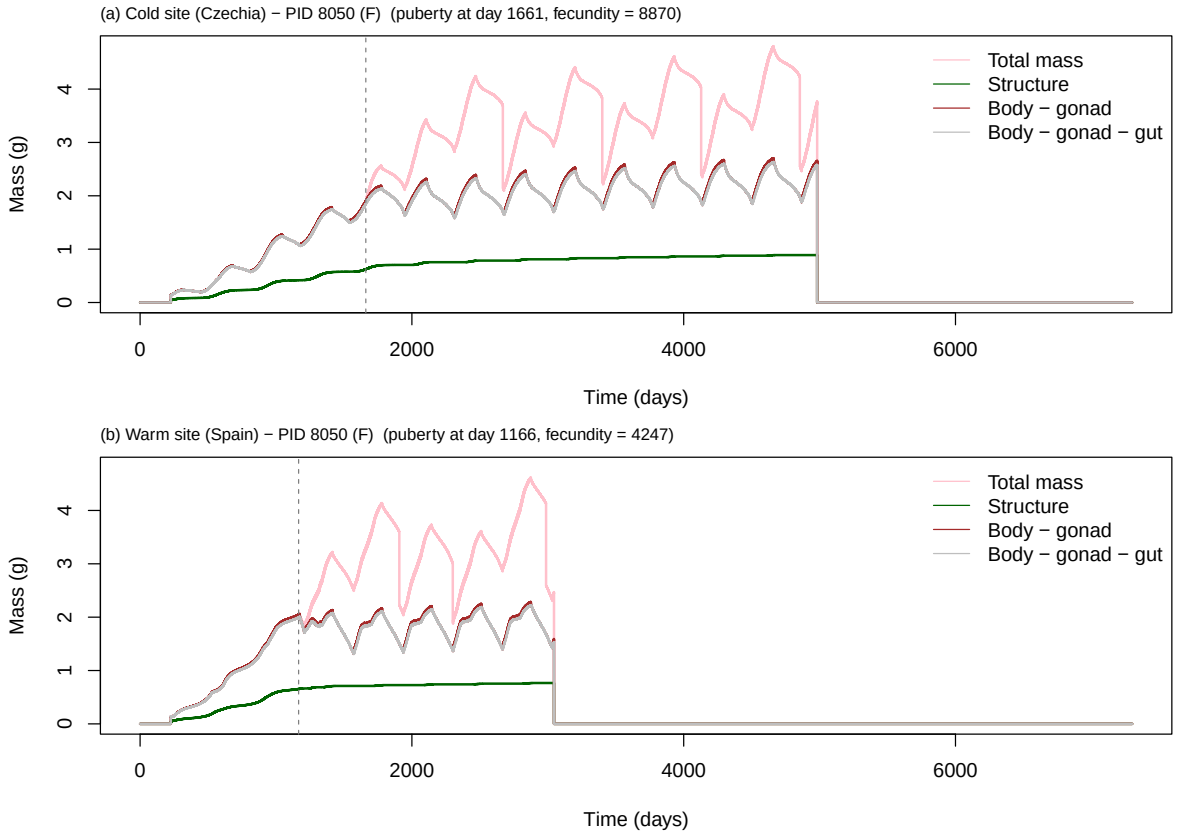


**Figure S2.7:** Growth trajectory for PID 8050 (female).

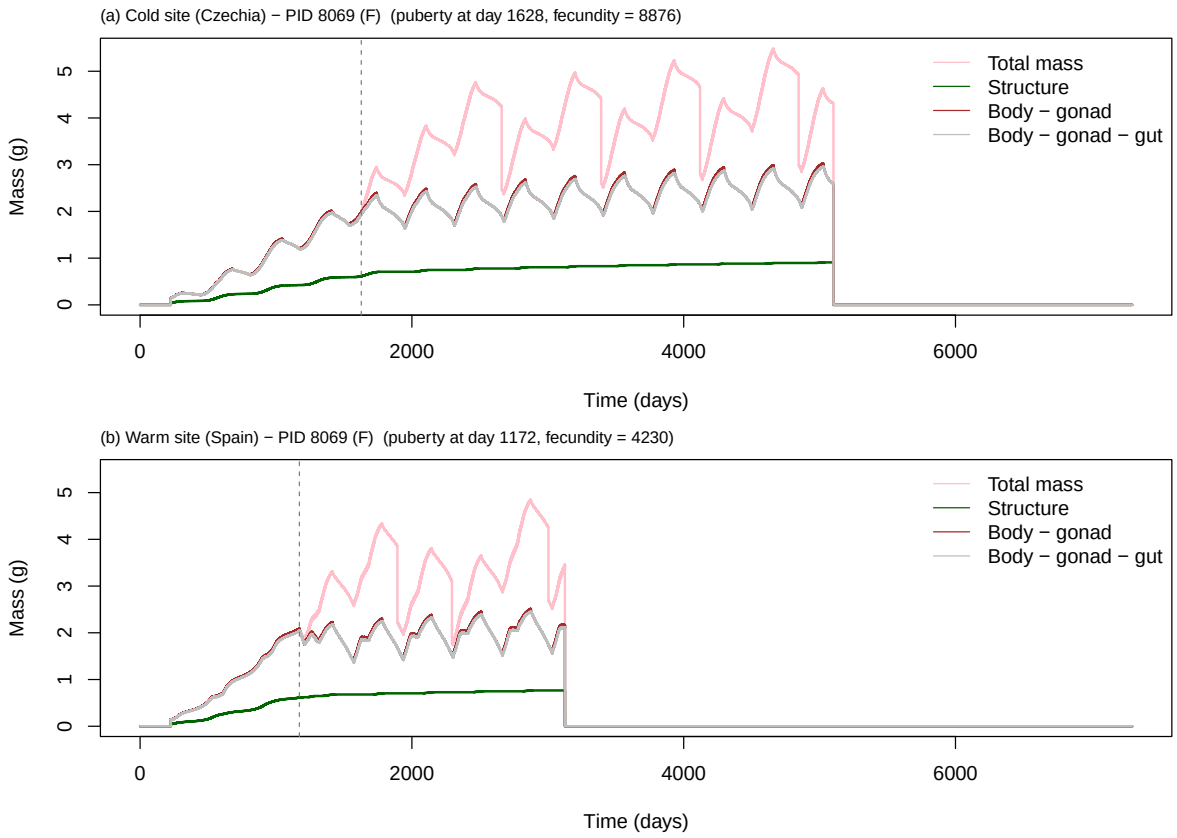


**Figure S2.8:** Growth trajectory for PID 8069 (female).

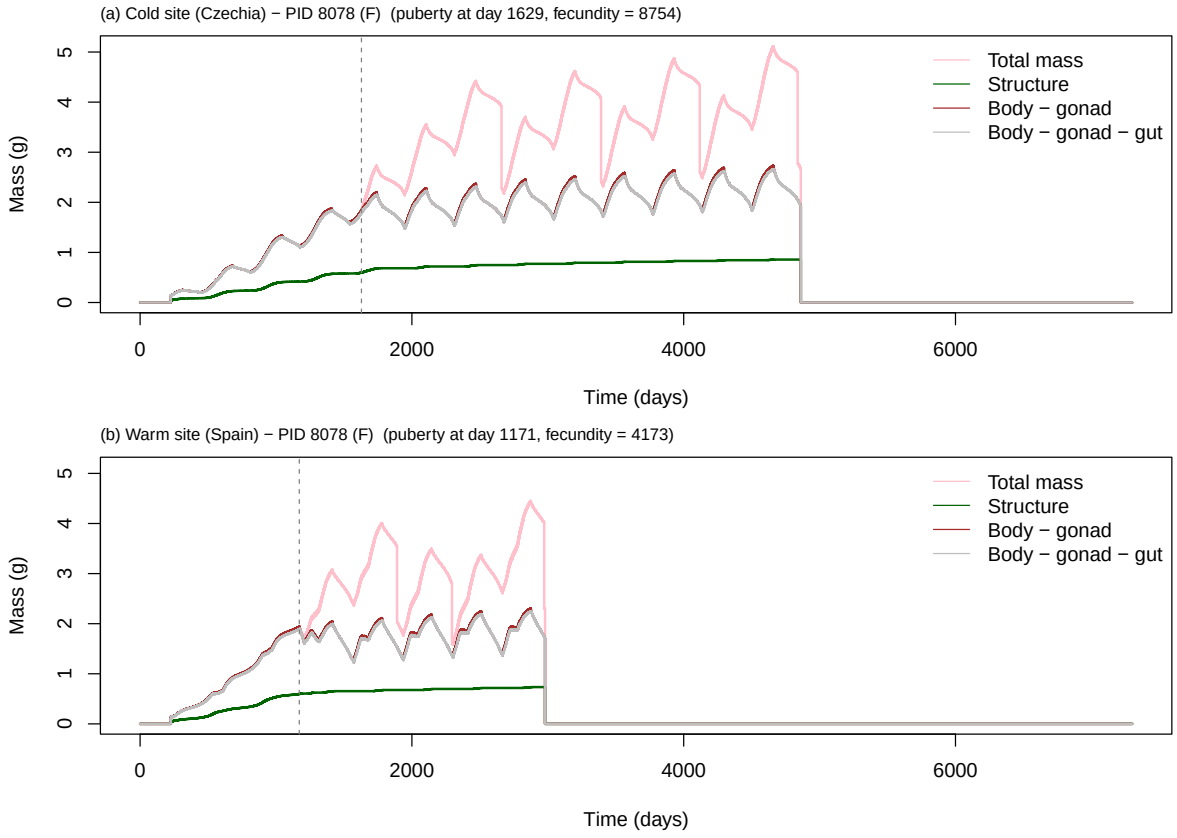


**Figure S2.9:** Growth trajectory for PID 8078 (female).

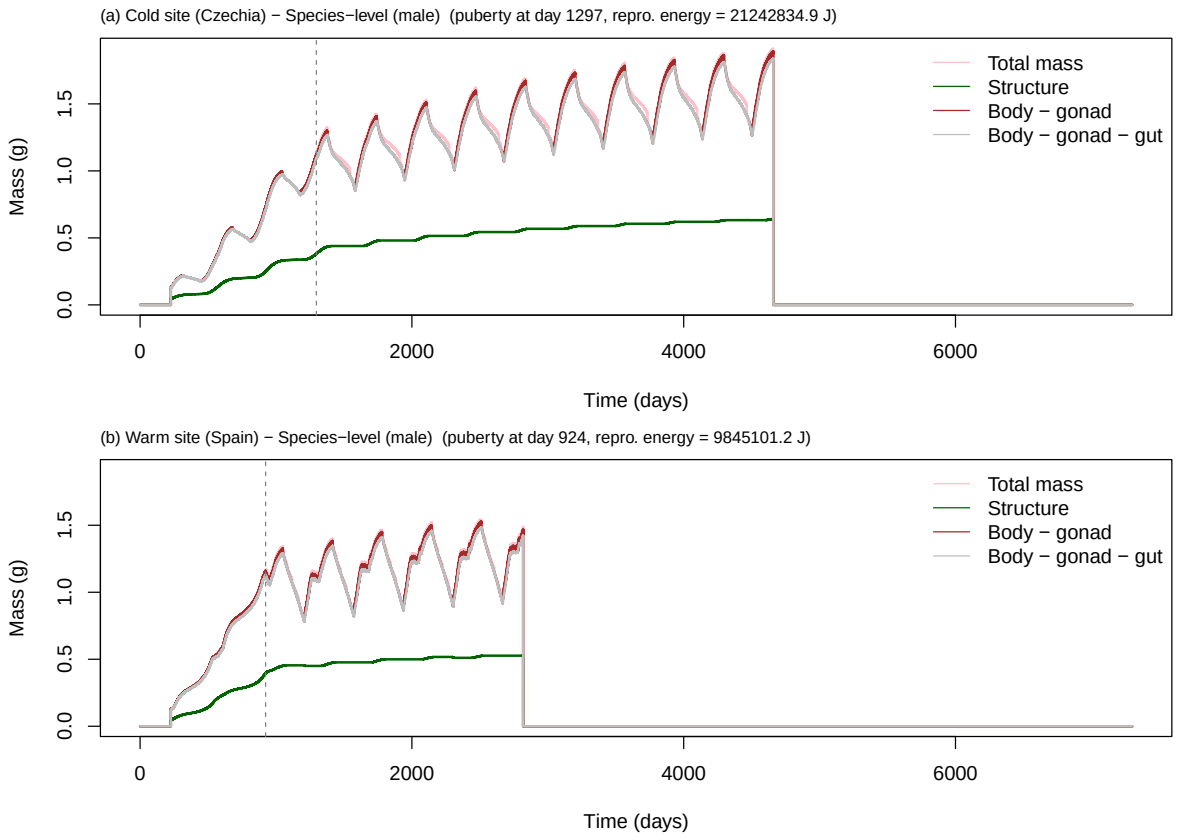


**Figure S2.10:** Growth trajectory for Species-level (male).

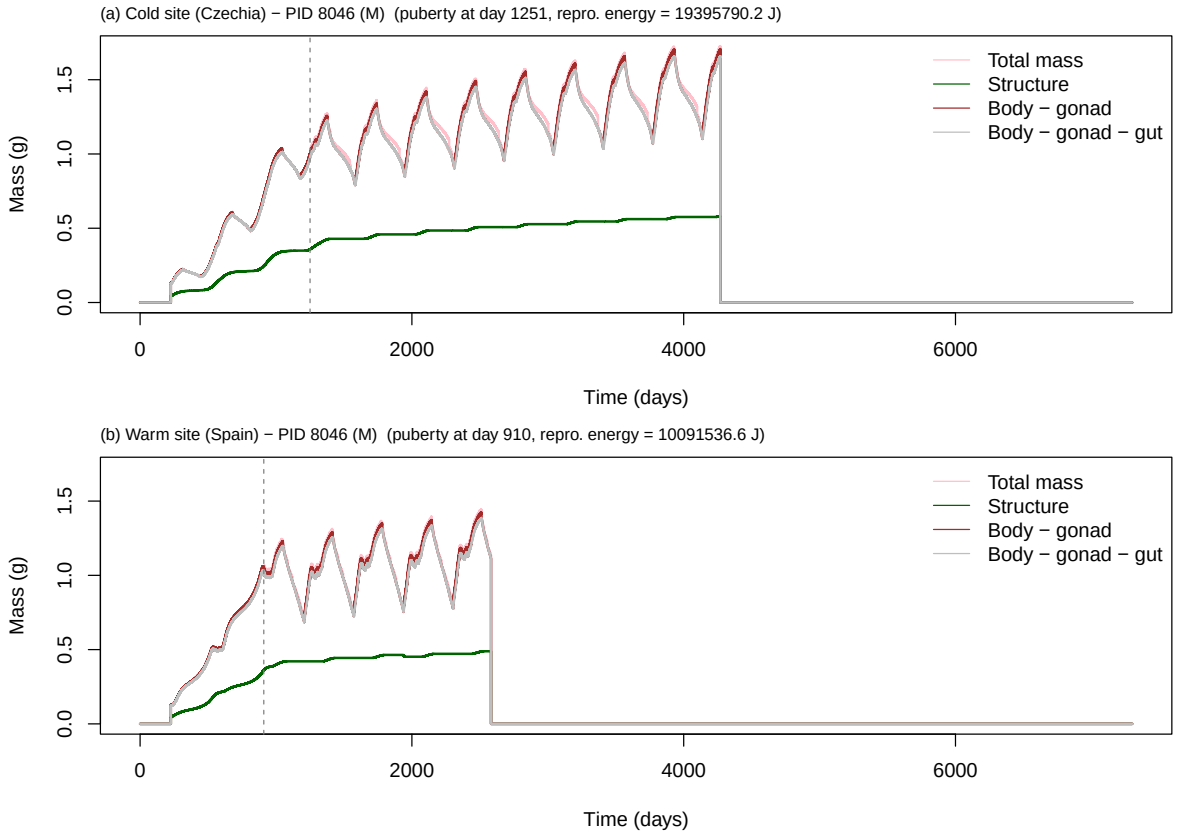


**Figure S2.11:** Growth trajectory for PID 8046 (male).

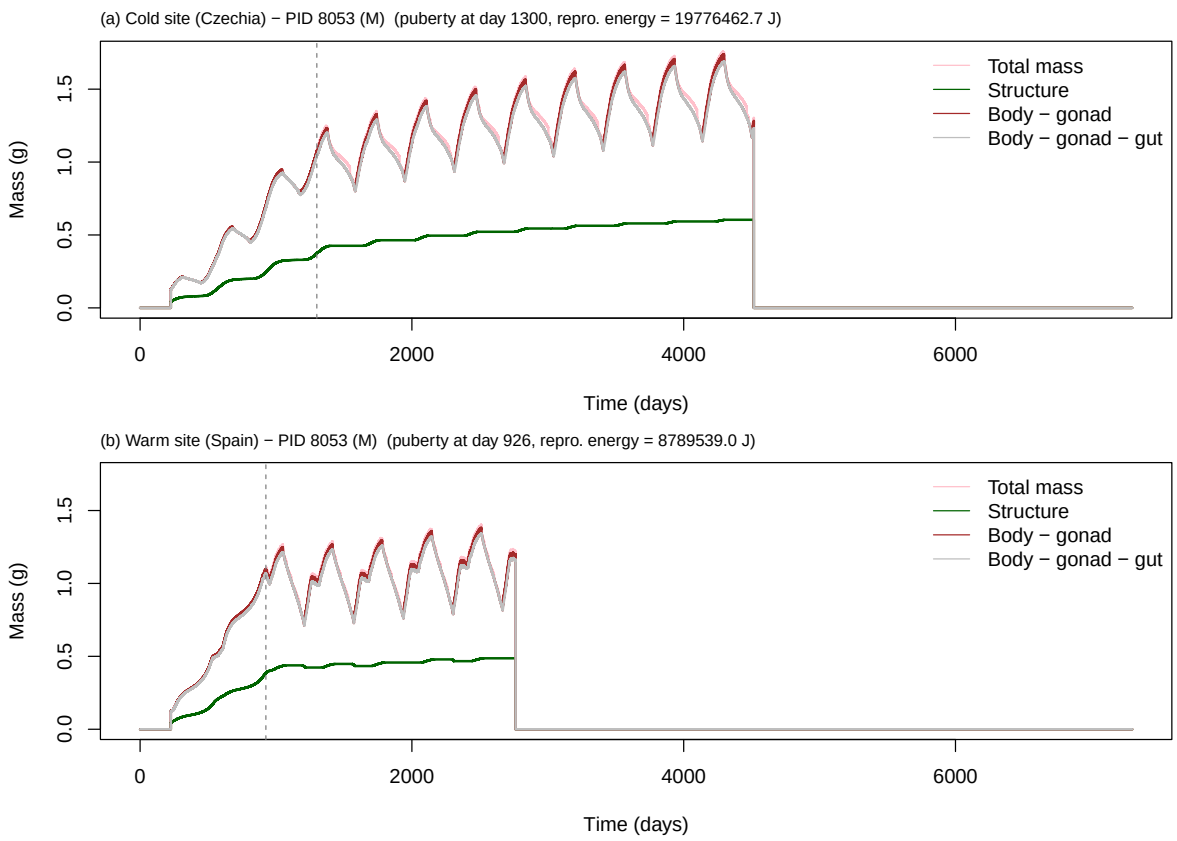


**Figure S2.12:** Growth trajectory for PID 8053 (male).

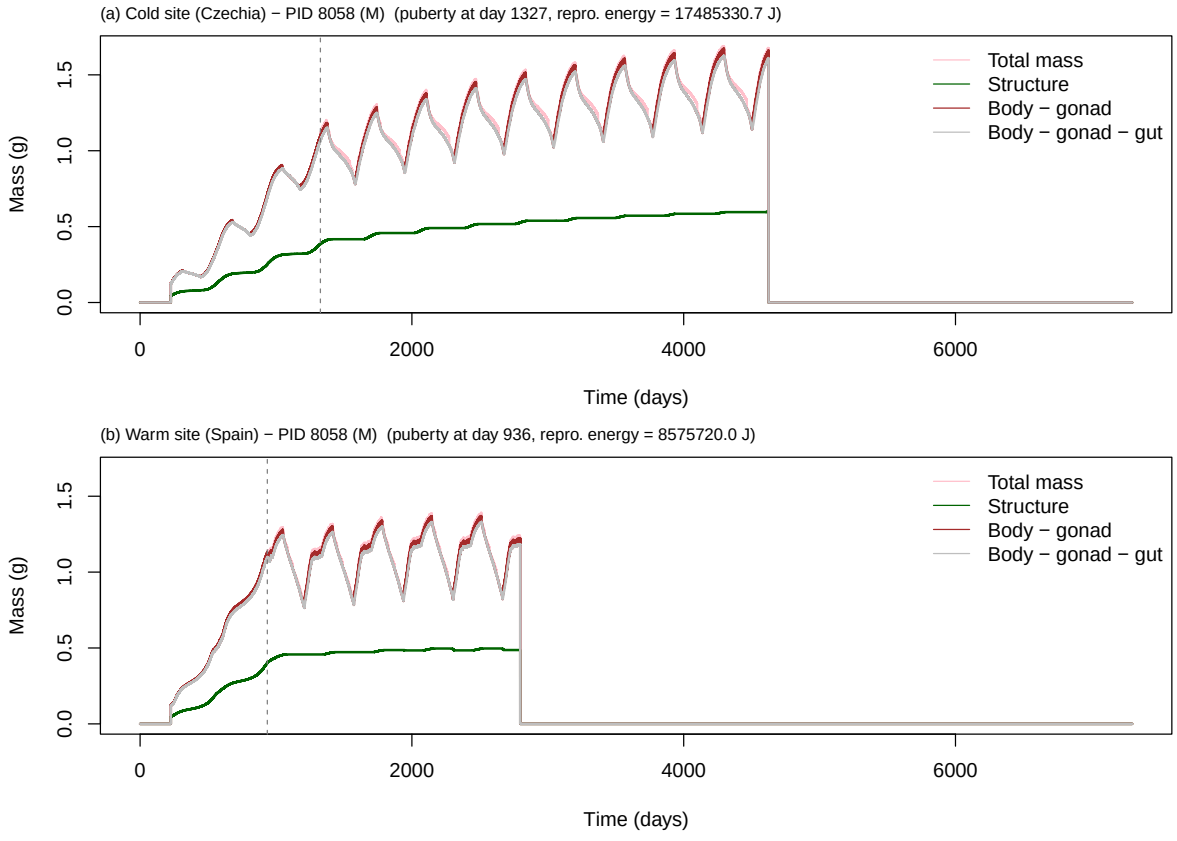


**Figure S2.13:** Growth trajectory for PID 8058 (male).

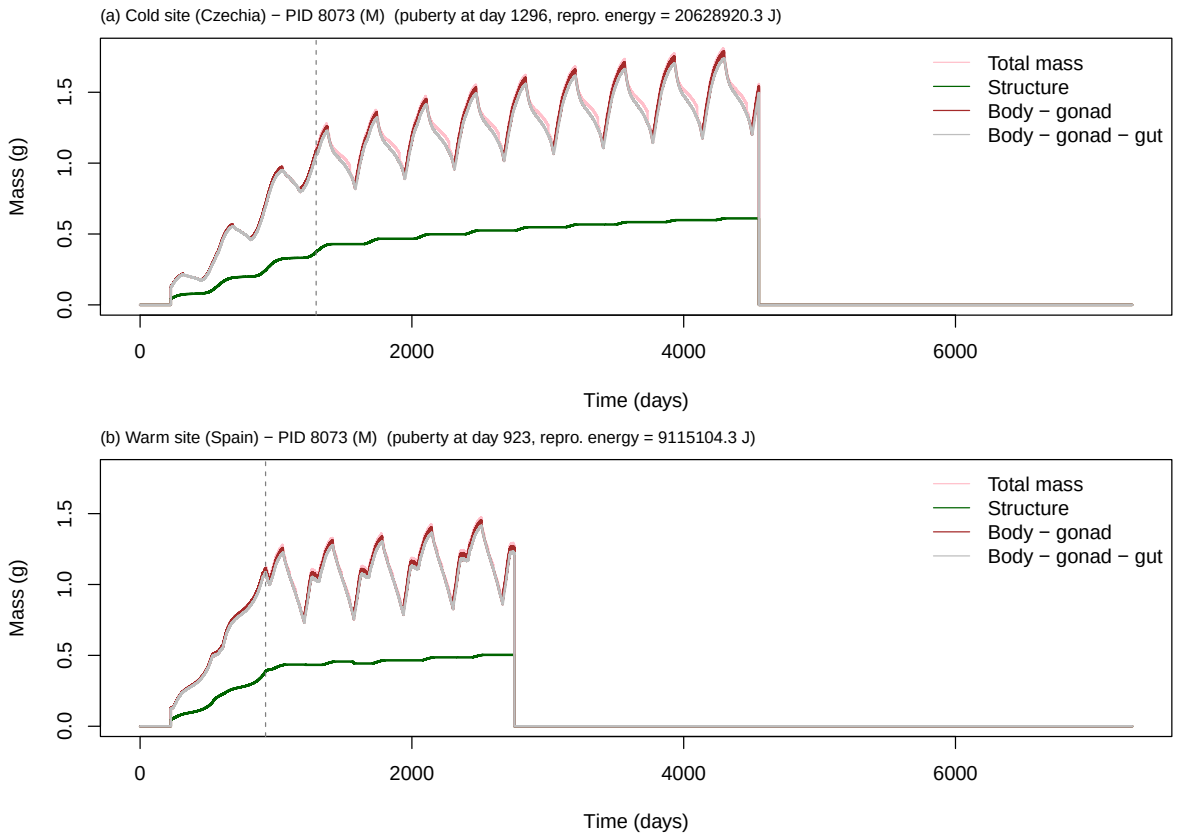


**Figure S2.14:** Growth trajectory for PID 8073 (male).

# 4 References

Kooijman, S. A. L. M. (2010). *Dynamic energy budget theory for metabolic organisation* (3rd ed.). Cambridge university press.

Marques, G. M., Augustine, S., Lika, K., Pecquerie, L., Domingos, T., & Kooijman, S. A. L. M. (2018). The AmP project: Comparing species on the basis of dynamic energy budget parameters (D. A. Beard, Ed.). *PLOS Computational Biology*, *14*(5), e1006100. https://doi.org/10.1371/journal.pcbi.1006100

# **Appendix 3** *for:* Beyond the average phenotype: individual variation alters mechanistic forecasts of climate responses

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# Additional Results

## 1 Jensen bias for males

~~Figure S3.1~~ **Figure S3.1** shows the spatial distribution of Jensen's dissimilarity index ($J$) between species-level and individual-level prediction surfaces for males under current European climate. While the species-level DEB parameterisation underestimated most performance metrics for females (**Fig. 2**), the pattern for males was largely opposite: overestimation dominated for body size, reserve, food, reproductive energy and basking metrics, whereas maximum and minimum body temperature showed the reverse pattern (Briscoe et al., 2023). Foraging time was more balanced, with similar proportions of positive and negative bias across the range. The magnitude and direction of Jensen bias therefore depended strongly on both sex and ecological performance metric.

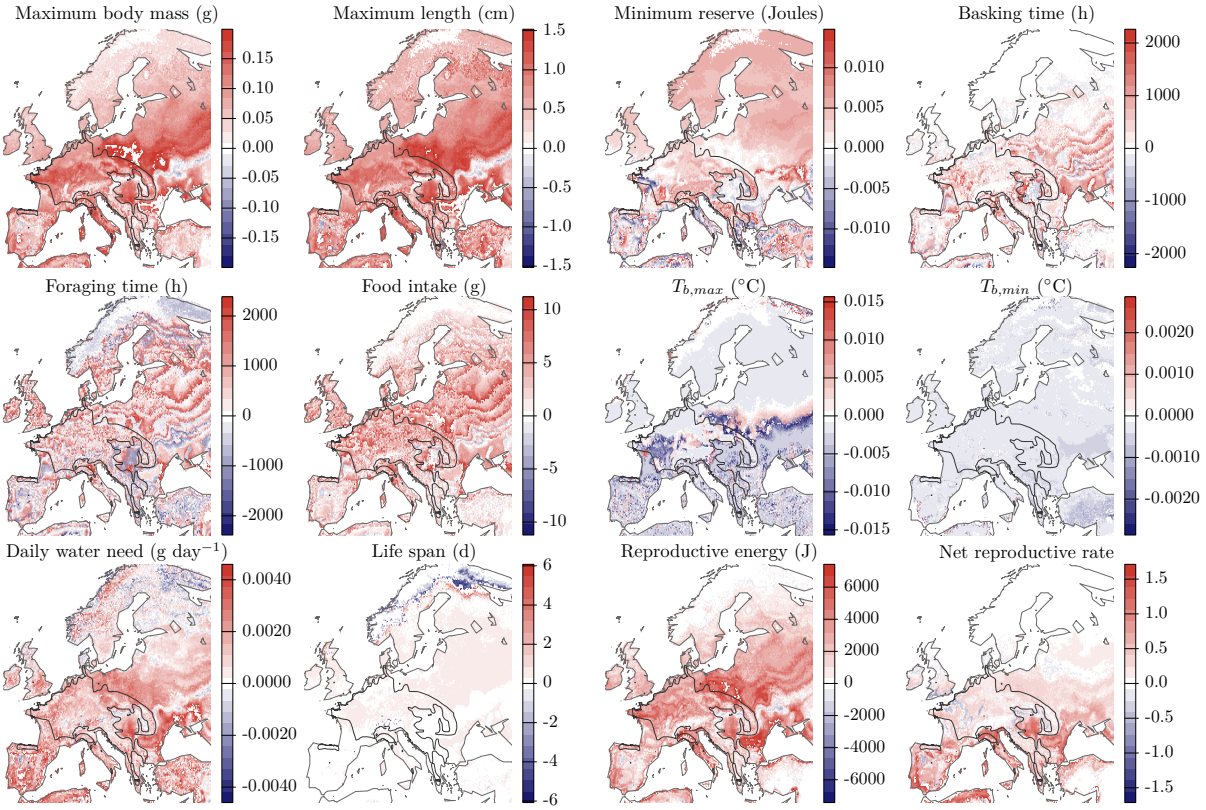


**Figure S3.1:** Jensen's bias ($J$) between species-level and individual-level prediction surfaces for males. Values close to zero indicate close agreement; departures from zero indicate systematic differences in the spatial pattern. The colour scale is symmetric and centred at zero.

# 2 Coefficient of variation maps

The coefficient of variation (~~CV~~CV) across the 37 individual-level predictions quantifies the degree of physiological heterogeneity independent of Jensen bias. High ~~CV~~ CV values identify regions and metrics where individual variation most strongly influences predicted outcomes.

## 2.1 Females

For females, ~~CV~~ CV was highest toward the range margins for most performance metrics (**Fig. S3.2**), indicating that the ecological consequences of individual physiological variation were spatially structured rather than uniform across the landscape.

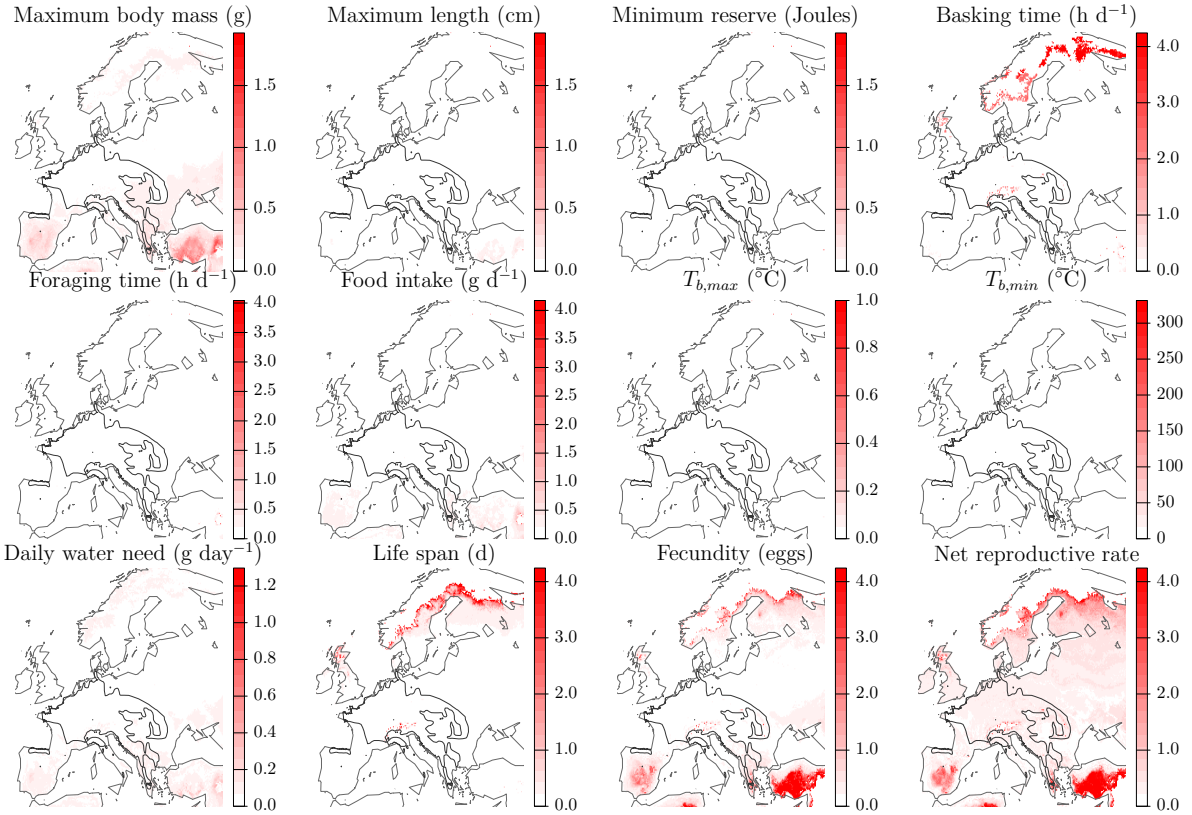


**Figure S3.2:** Coefficient of variation (~~CV~~CV) across individual-level predictions for females. Each panel shows the spatial distribution of ~~CV~~ CV for one performance metric. Higher values indicate greater individual-level variation in that metric across the geographic range.

## 2.2 Males

A similar spatial pattern was observed for males (**Fig. S3.3**), with ~~CV~~ CV values peaking at range margins for most metrics, consistent with the pattern observed in females.

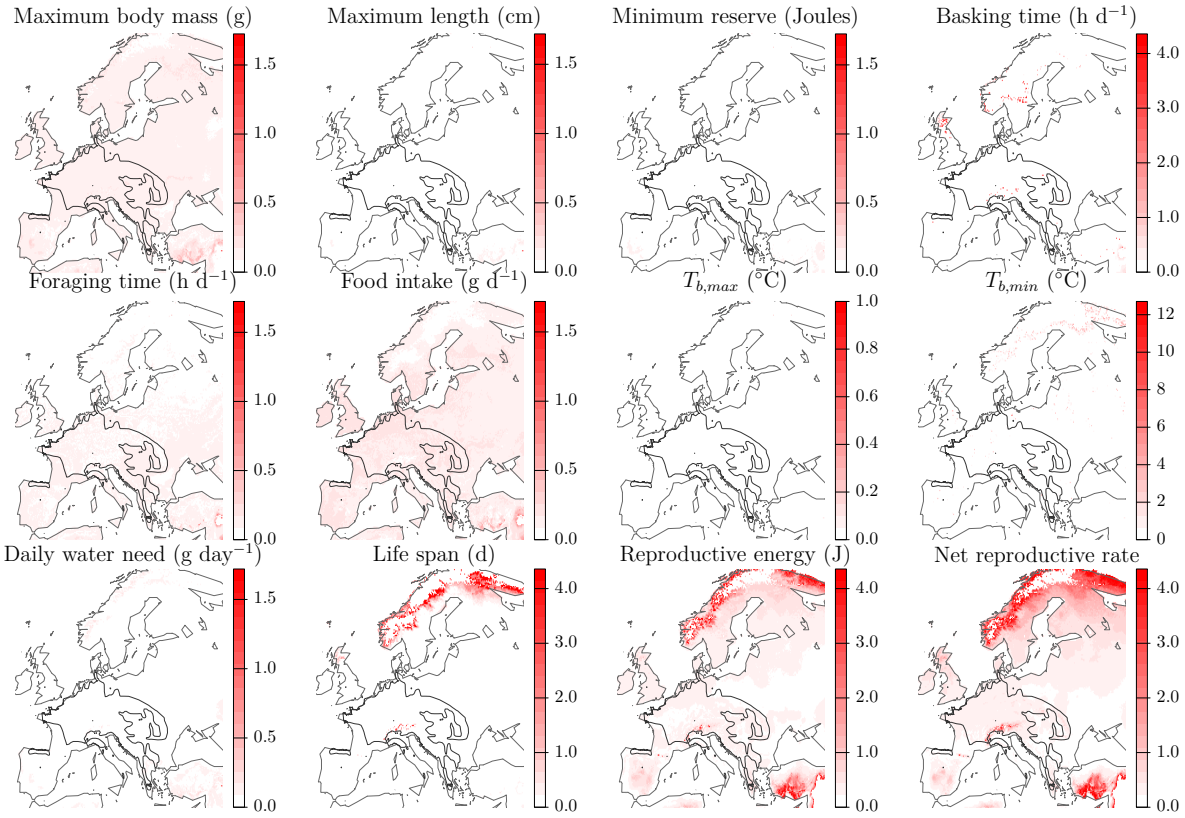


**Figure S3.3:** Coefficient of variation ($CV$) across individual-level predictions for males. Each panel shows the spatial distribution of $CV$ for one performance metric. Higher values indicate greater individual-level variation in that metric across the geographic range.

# 3 Climate warming reshapes Jensen bias (males)

Climate warming altered the magnitude of Jensen bias for all twelve male performance metrics (**Fig. S3.4**), mirroring the pattern observed for females (**Fig. 3**). However, as for females, warming did not produce a uniform response: the strongest amplification occurred for reproductive energy and basking time, whereas maximum and minimum body temperature were attenuated. The sign of $J$ changed in a substantial proportion of cells under +4 °C warming, with highest rates for basking time, foraging time and lifetime fecundity. Even under +2 °C warming, the direction of change was already discernible for most metrics, with eleven of twelve metrics showing increased $|J|$ and two metrics switching from amplification to attenuation at +4 °C, suggesting a nonlinear threshold in the species-averaging bias (Ingram et al., 2018).

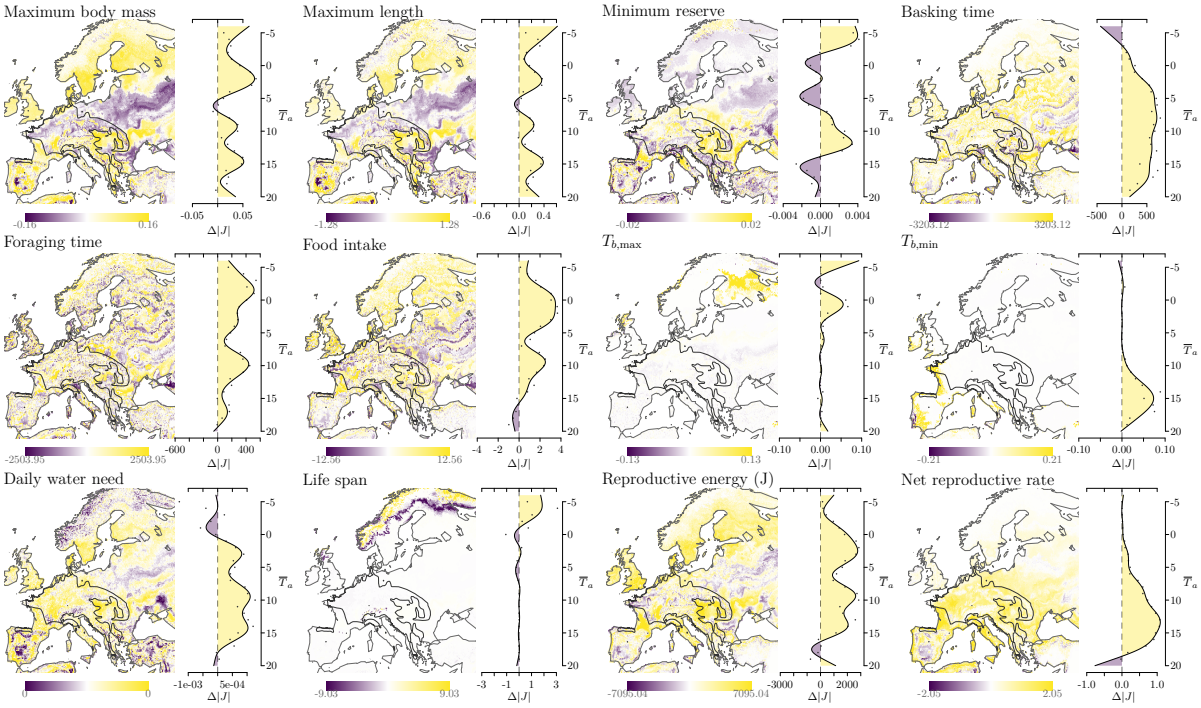


**Figure S3.4:** Jensen's bias amplification under +4 °C warming for males. Each map displays the change in absolute Jensen dissimilarity ($\Delta|J|$, left, with colour bar) for each performance metric. The curve (GAM smoother and 95 % confidence band) at the right of each map represents how $\Delta|J|$ changes along the mean annual temperature ($\overline{T}_a$) gradient. *Yellow* shading (positive $\Delta|J|$) indicates amplification of the species-averaging bias under warming, while *purple* shading (negative $\Delta|J|$) indicates attenuation.

# 4 Consequences for ecological forecasts (males)

Jensen bias produced trait-specific differences between species-level and individual-level forecasts for males under both current and future climates (**Fig. S3.5**), mirroring the female pattern (**Fig. 4**). Reproductive energy exhibited the largest discrepancies, with the species-level model generally overpredicting relative to the individual-level mean under the current climate. Lifespan showed a consistent tendency for the species-level model to predict higher values than the individual-level mean across climate scenarios, whereas maximum body mass showed comparatively close agreement, indicating relatively weak consequences of physiological averaging for this trait (Wolf & McKechnie, 2023). The relative bias maps (**Figs. S3.6, S3.7**) express the same comparison as a percentage of the individual-level mean, making the spatial structure of the disagreement visible.

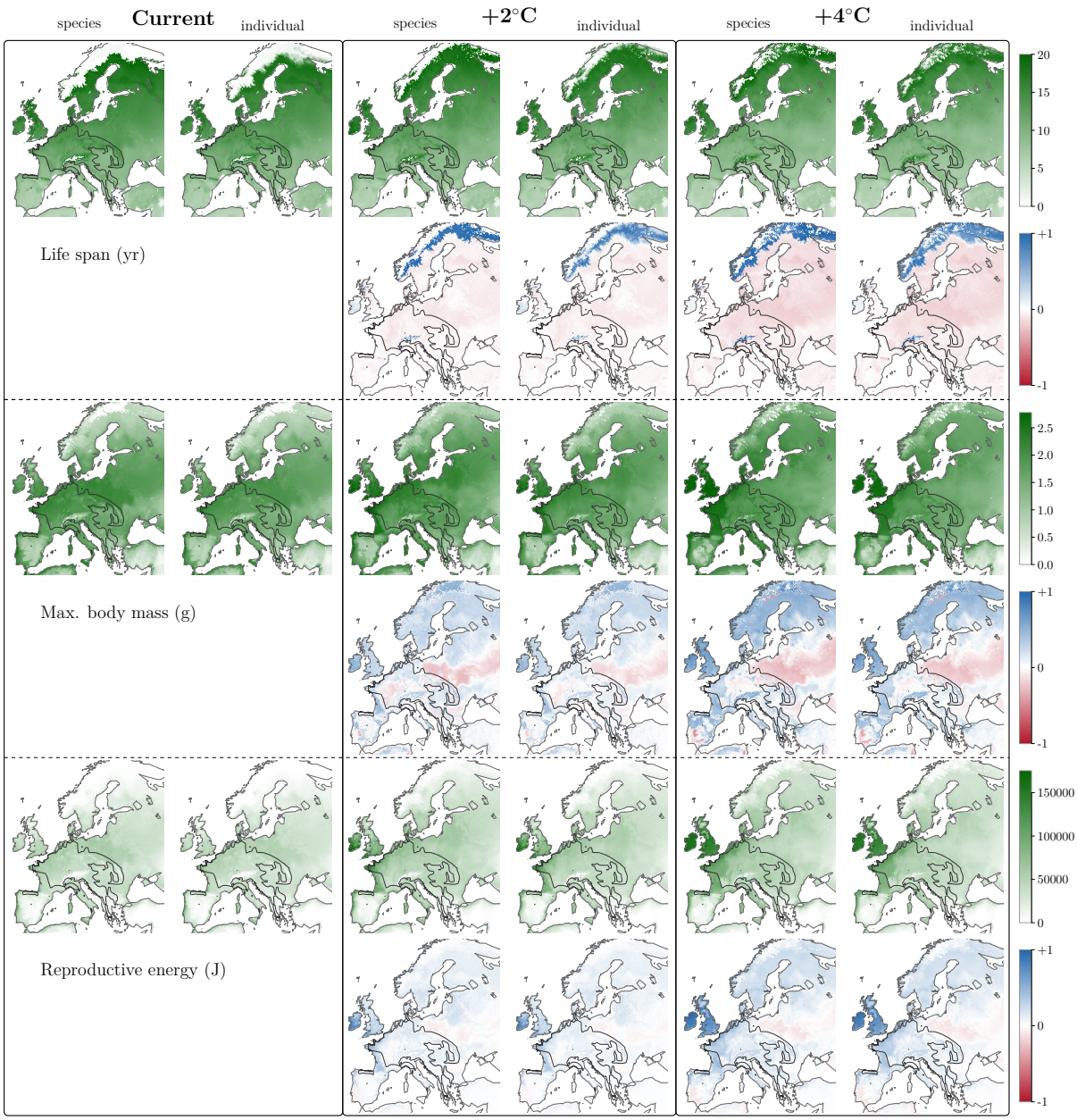


**Figure S3.5:** Comparison of species-level and mean individual-level predictions for life span, maximum body mass and reproductive energy for males under current, +2 °C and +4 °C climate scenarios. For each metric, the top row shows the raw predicted values for all six parameterization–climate combinations on a shared scale. At the bottom, we show the change relative to current climates for the warming scenarios ($P_{\text{future}} - P_{\text{current}}$) on a normalized scale $[-1, 1]$ (*blue* indicates decrease, while *red* indicates increase).

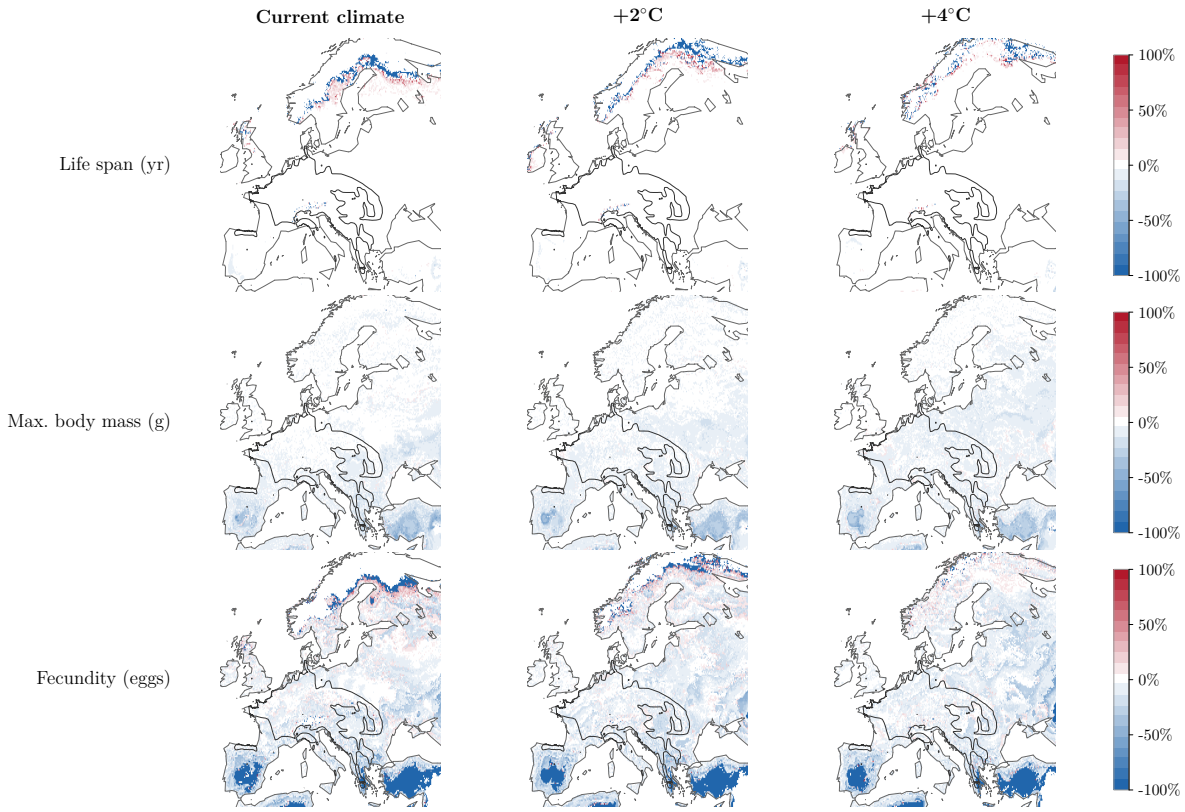


**Figure S3.6:** Relative bias (%) between species-level and individual-level predictions for females across current, +2 °C and +4 °C scenarios. Relative bias is computed as $R = (P_{\text{species}} - \overline{P}_{\text{ind}})/|\overline{P}_{\text{ind}}| \times 100$ for each grid cell. Blue indicates underestimation by the species-level model ($R < 0$); red indicates overestimation ($R > 0$); white indicates close agreement ($R \approx 0$).

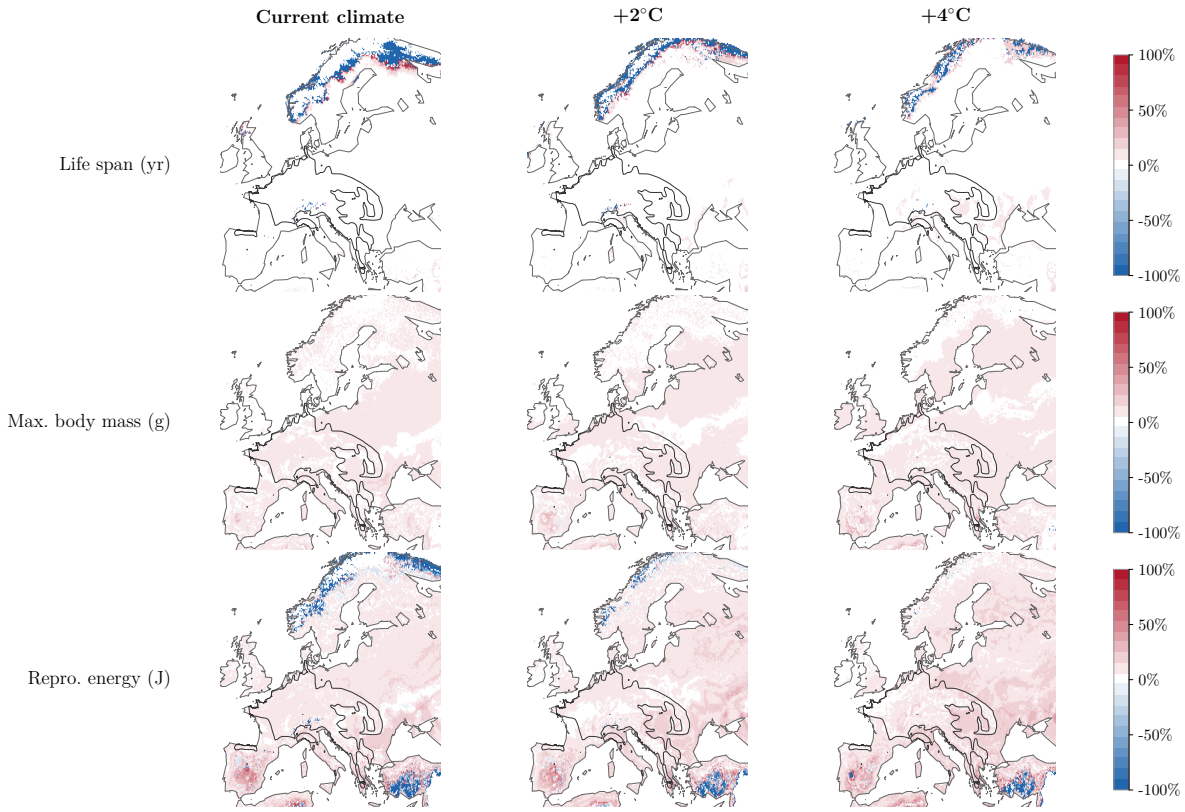


**Figure S3.7:** Relative bias (%) between species-level and individual-level predictions for males, as in **Fig. S3.6** but showing lifespan, maximum body mass and reproductive energy.

# 5 References

Briscoe, N. J., Morris, S. D., Mathewson, P. D., Buckley, L. B., Jusup, M., Levy, O., Maclean, I. M. D., Pincebourde, S., Riddell, E. A., Roberts, J. A., Schouten, R., Sears, M. W., & Kearney, M. R. (2023). Mechanistic forecasts of species responses to climate change: The promise of biophysical ecology. *Global Change Biology*, *29*(6), 1451–1470. https://doi.org/10.1111/gcb.16557

Ingram, T., Costa-Pereira, R., & Araújo, M. S. (2018). The dimensionality of individual niche variation. *Ecology*, *99*(3), 536–549. https://doi.org/10.1002/ecy.2129

Wolf, B. O., & McKechnie, A. E. (2023). Biophysical approaches to predicting species vulnerability. *Global Change Biology*, *29*(6), 1421–1422. https://doi.org/10.1111/gcb.16565